

Week 4

Genomics — Sequencing, Variant Interpretation, and AI-Assisted Analysis

GEO, NCBI, and Reproducible R Workflows

Bioinformatics: From Multi-Omics Data to Discovery

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Two 40-Minute Sessions | Fall 2026

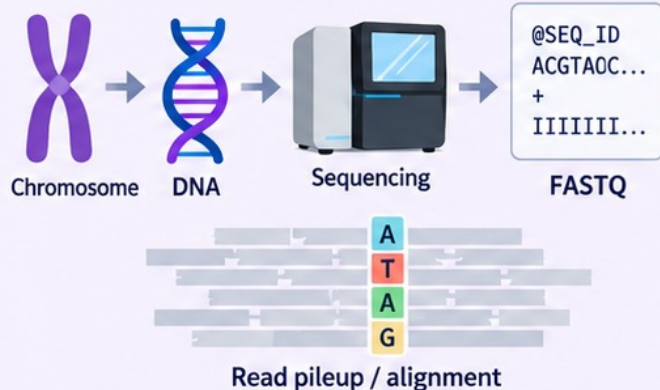
Today's Genomic Omics Roadmap

Three connected parts: measurement, interpretation, and AI-supported reproducibility.

1

Measurement

Genome to data



- Reference genome
- Sequencing platforms
- FASTQ and quality
- Alignment and variant evidence

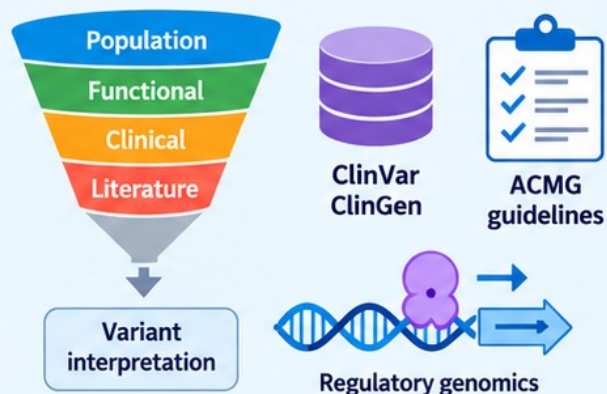


Activity: rank read pileups

2

Interpretation

Pipeline to claim



- Variant classes
- Population and clinical evidence
- Regulatory genomics
- From evidence to claim

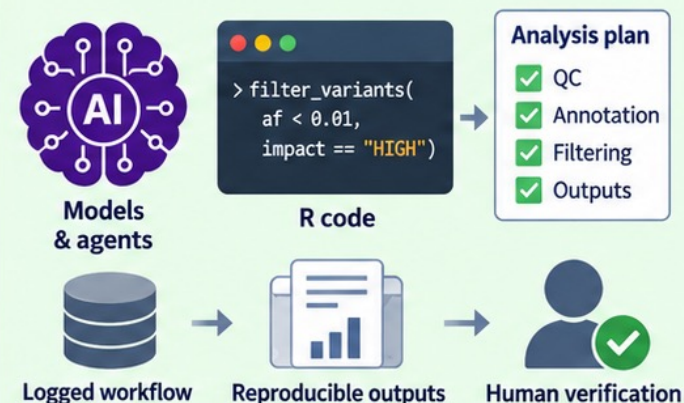


Activity: compare variant evidence

3

AI + Reproducibility

Plan to decision



- AI-assisted QC and annotation
- Inspectable workflows
- Plan-first prompts
- Human verification



Activity: audit AI-generated code



DNA/omics



Pipeline



Interpretation



AI-assisted verification



Take-home:

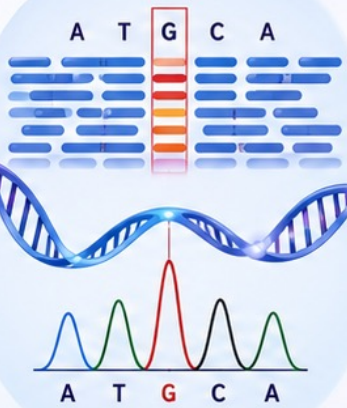
measurement + interpretation + AI

The genomic omics learning pathway

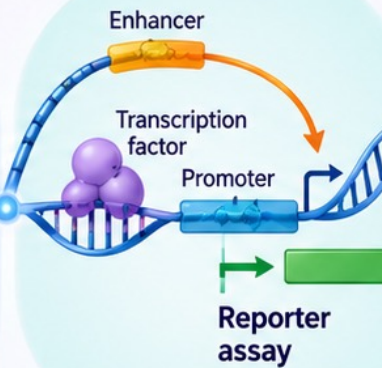
1 DNA foundations



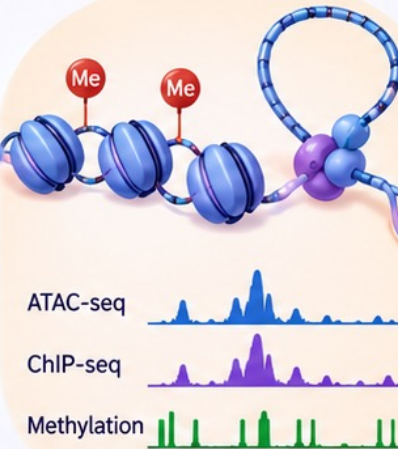
2 Sequencing and variants



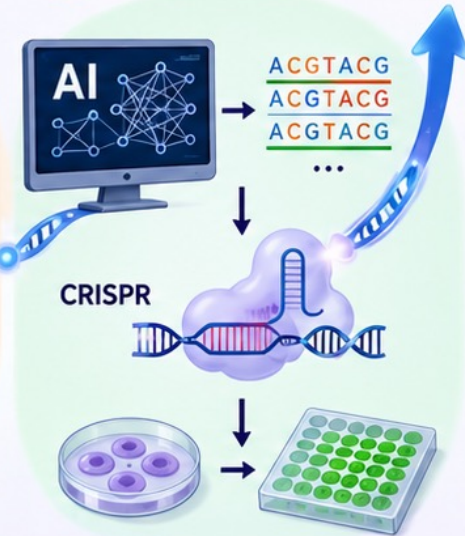
3 Regulatory DNA



4 Chromatin and 3D genome



5 AI-guided experiments



Dry lab

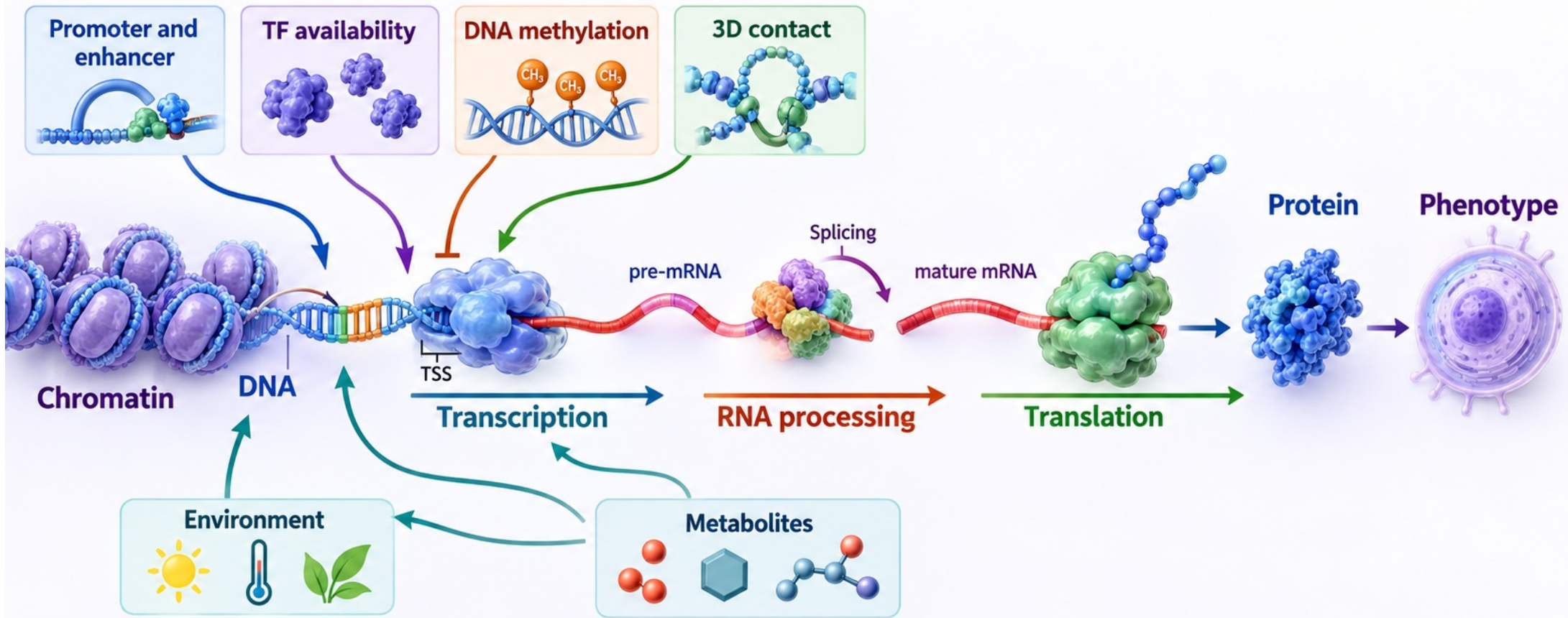


Questions refine
the next experiment

Wet lab



The central dogma begins with regulated DNA

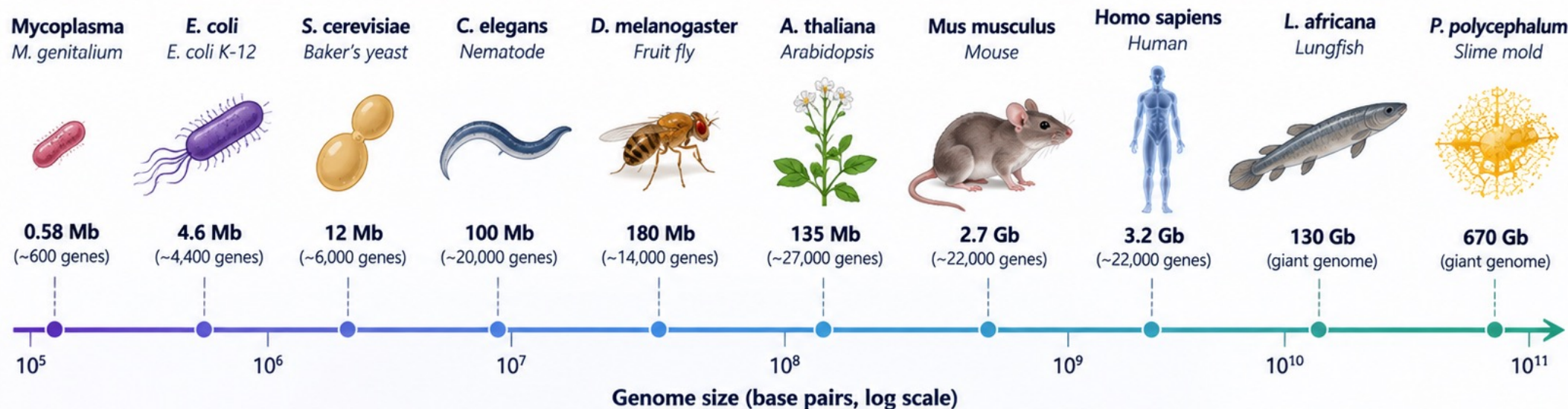


Sequence sets capacity. Regulatory state sets context.

Genome size spans several orders of magnitude

From compact bacterial genomes to large and complex eukaryotic genomes

SAME
INFORMATION
DIFFERENT
SCALE



Prokaryotes

Eukaryotes

Giant genomes



$\sim 10^6$ bp

Typical bacterial genomes are 1–10 million base pairs.



$\sim 10^7$ – 10^9 bp

Most eukaryotic genomes range from 10 million to several billion base pairs.



$> 10^{10}$ bp

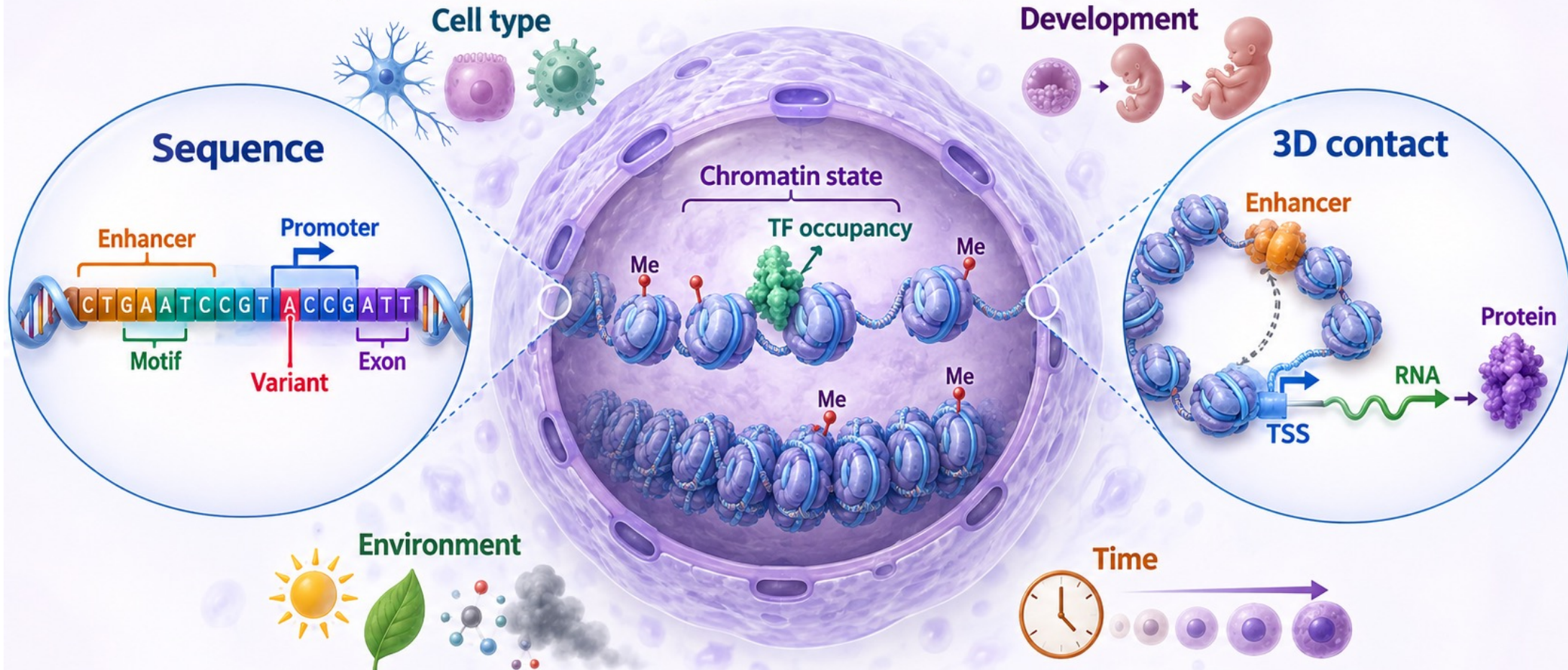
Some species (e.g., lungfish, slime mold) have extremely large genomes.



Genome size $\neq$ gene number

Larger genomes often contain more non-coding and repetitive sequences.

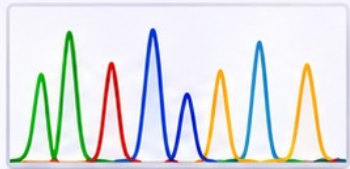
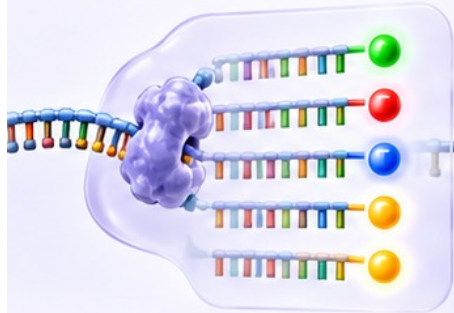
DNA is a sequence and a regulated physical system



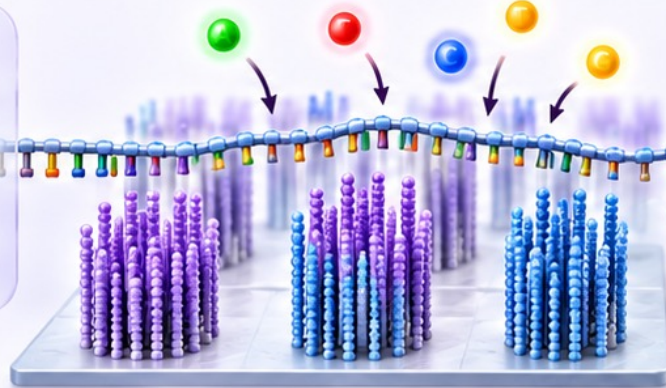
Identical DNA can support different expression programs.

Sequencing technologies answer different questions

Sanger
Chain termination



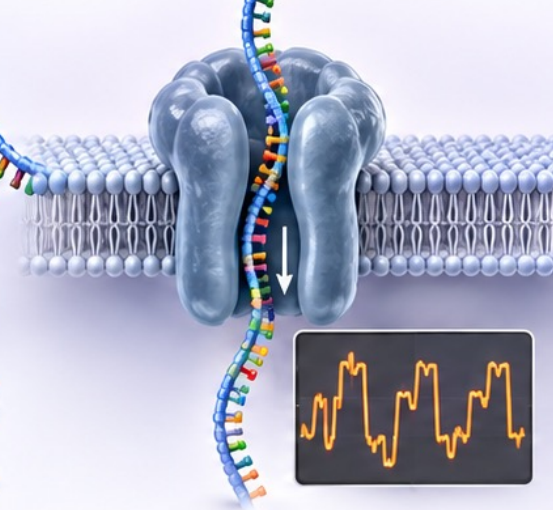
Illumina
Cyclic incorporation



PacBio HiFi
Circular consensus



Oxford Nanopore
Electrical current



Short reads

Long reads

SNVs and small indels

WES and WGS

Phasing and repeats

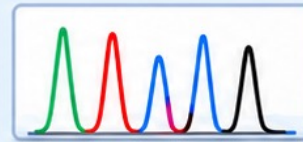
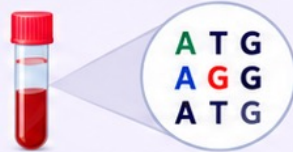
Structural variants

Read length, accuracy, depth, and cost define the observable variant space.

No platform is universally best – the biological question chooses the assay

A

Confirm one known SNV in a clinic sample



Sanger

Cheap, exact, one amplicon.

B

Exomes of 10,000 population samples



Illumina

Throughput drives cost down.

C

Suspected Huntington repeat expansion

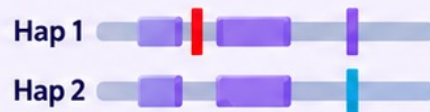


PacBio HiFi / ONT

Reads must span the repeat.

D

Phase variants across a whole gene



Long reads

One molecule, one haplotype.

E

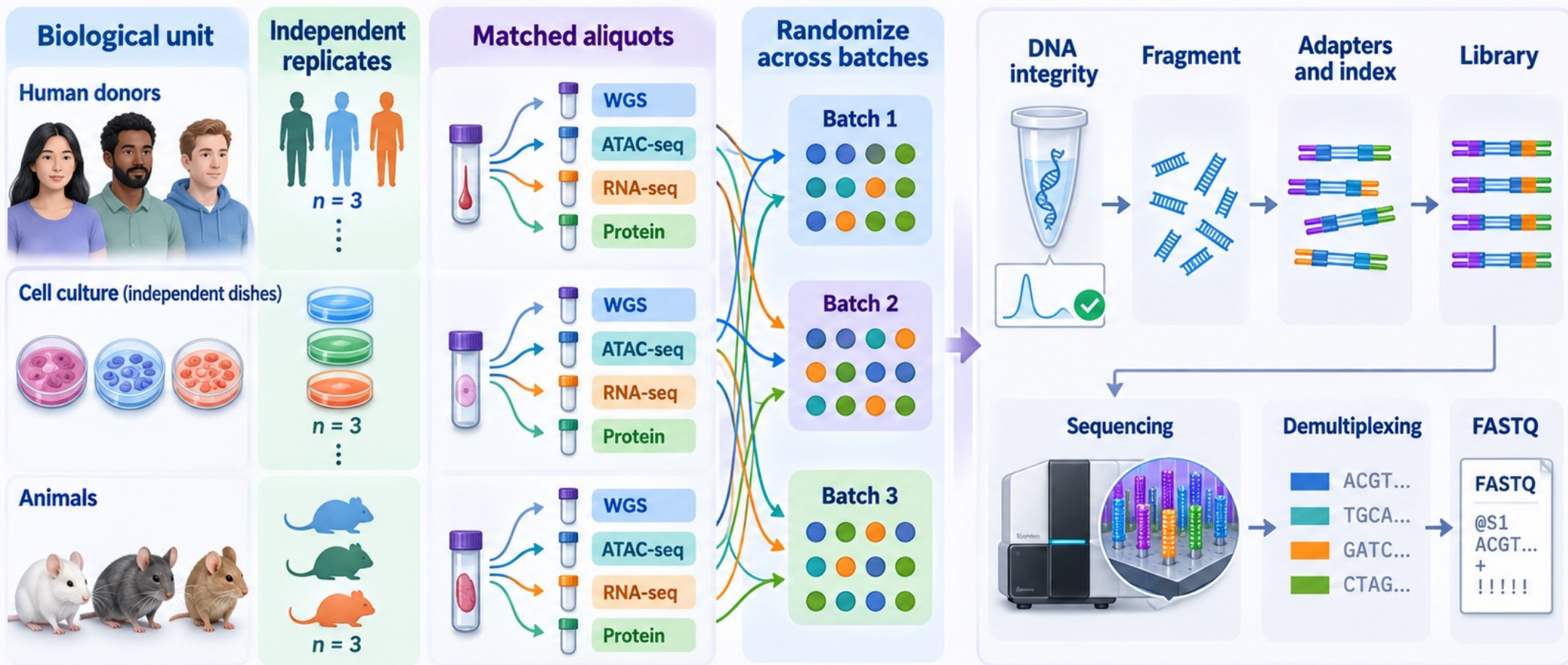
Outbreak surveillance in the field



Oxford Nanopore

Portable, real-time.

Study design determines what sequencing can reveal

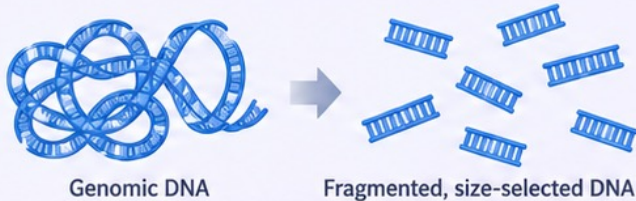


Technical repeats do not replace biological replication.

Illumina sequencing converts an indexed library into reads plus base qualities

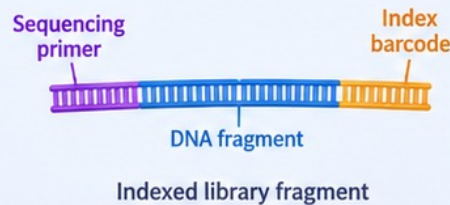
1 Fragment DNA

Break genomic DNA into short, size-selected pieces.



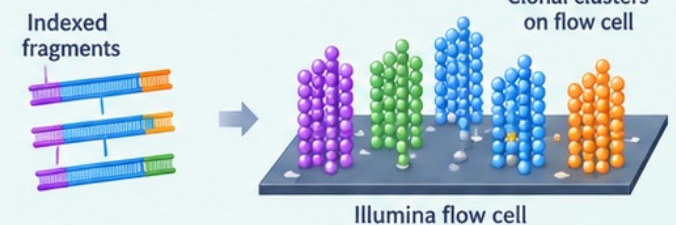
2 Ligate Adapters

Add sequencing primers and sample-specific index barcodes.



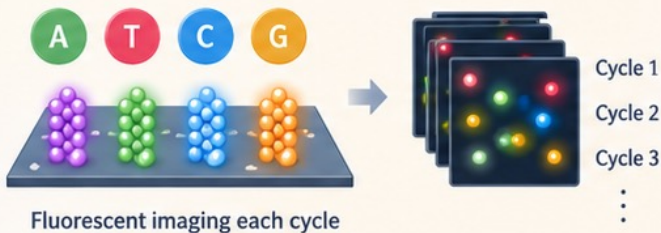
3 Cluster Generation

Each fragment is amplified into a clonal cluster on the flow cell.



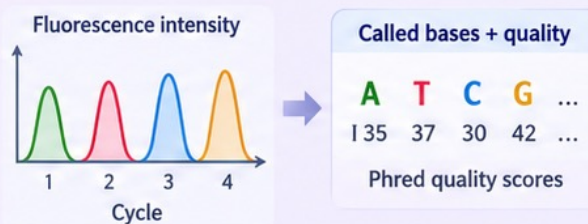
4 Sequence by Synthesis

One fluorescent base incorporated per cycle, imaged across millions of clusters.



5 Base Calling

Signal intensities become bases plus per-base quality scores.



6 Demultiplex + FASTQ

Reads are sorted by index barcode and written as FASTQ files.



Bias points: fragmentation, PCR amplification, and cluster chemistry all leave signatures you will meet again in QC.



Output of the wet lab + instrument: millions of reads, each with a quality string — the FASTQ file.

A FASTQ record stores sequence and confidence, not biological interpretation

```
1 @SEQ-ID:flowcell:lane:1101:1724:2108
2 GATTGTTGGGTTTCAAAGCAGTATCGATCAAATAG
3 +
4 !' '*((( (***) )%%%)++)(%%%) . 1***-+*''
```



What is NOT in a FASTQ: where the read came from, whether it maps uniquely, or what variant it might support.

All of that is computed *later* — by alignment and calling.

1

Line 1 — Header:

instrument, flow cell, lane, tile coordinates, read pair info.

2

Line 2 — Sequence:

the observed bases, exactly as the instrument called them.

3

Line 3 — Separator:

“+”, optionally repeating the header.

4

Line 4 — Quality string:

one ASCII character per base, encoding its Phred score.

Phred scores convert base-calling uncertainty into an error probability

Phred quality score (Q)

$$Q = -10 \log_{10}(P_{\text{error}})$$

Higher Q → lower error probability
(10× increase in Q = 10× lower error rate)

Q Score	P(error)	Base call accuracy
Q10	1 in 10 (0.1)	90%
Q20	1 in 100 (0.01)	99%
Q30	1 in 1,000 (0.001)	99.9%
Q40	1 in 10,000 (0.0001)	99.99%

Visualizing quality



A FASTQ record stores sequence and confidence

```
@SEQ_ID:flowcell:lane:1101:1724:2108
GATTTGGGGTTCAAAGCAGTATCGATCAAATAG
+
! '* ( ( ( ***+ ) ) % % + + ) ( % % % ) . 1 *** - + * ' '
```

Line 1 — Header
Instrument, flow cell, lane, tile coordinates, read pair info.

Line 2 — Sequence
Observed bases (A, C, G, T), exactly as called by the instrument.

Line 3 — Separator
"+", optionally repeating the header.

Line 4 — Quality string
One ASCII character per base, encoding its Phred score.

What do QC tools (e.g., FastQC) do for FASTQ data?



Quality control for raw sequencing reads

- ✓ **Per-base quality** Quality scores across read positions
- 📊 **Per-sequence quality** Distribution of mean quality
- 🅒 **N content** Proportion of ambiguous bases (N)
- 🔍 **Adapter content** Detection of adapter contamination
- 📋 **Sequence duplication** Overrepresented sequences
- ⚠ **Other metrics** GC content, sequence length, k-mer enrichment

→ QC helps assess data quality, detect technical issues, and decide on trimming/filtering before downstream analysis.



Calibration, not guarantee. Q30 means the *estimated* error rate is 0.1% — it does not prove this particular base is biologically reliable. That judgment needs read context (mapping, nearby variants, depth, etc.).

FastQC: Quality control for high-throughput sequencing data

A widely used tool to assess the quality of raw sequencing reads

FROM
DATA TO
DISCOVERY

 **Website** <https://www.bioinformatics.babraham.ac.uk/projects/fastqc/>

 **Software operational interface (Desktop version)** FastQC v0.12.1



Home Tools Courses Help About Us

FastQC

A quality control tool for high throughput sequence data

FastQC

- Introduction
- Downloads
- Documentation
- Release Notes
- More Information
- Citation

Introduction

FastQC is a quality control tool for high throughput sequence data. It provides a modular set of analyses which you can use to give a quick impression of whether your data has any problems before you do any further analysis.

[Download FastQC](#)

FastQC 0.12.1

File Tools Help

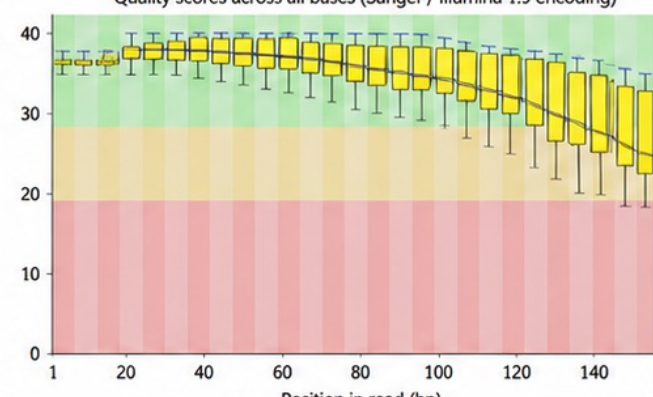
Open... Save... Quit

sample_R1.fastq.gz

- Basic Statistics
 - Per base sequence quality
 - Per tile sequence quality
 - Per sequence quality scores
 - Per base sequence content
 - Per sequence GC content
 - Per base N content
 - Sequence Length Distribution
 - Sequence Duplication Levels
 - Overrepresented sequences
 - Adapter Content

Per base sequence quality

Quality scores across all bases (Sanger / Illumina 1.9 encoding)



Position in read (bp)

Key features



Easy to use
GUI and command line



Multiple QC metrics
Assess sequencing quality from different aspects



Interactive reports
HTML output with plots and summaries



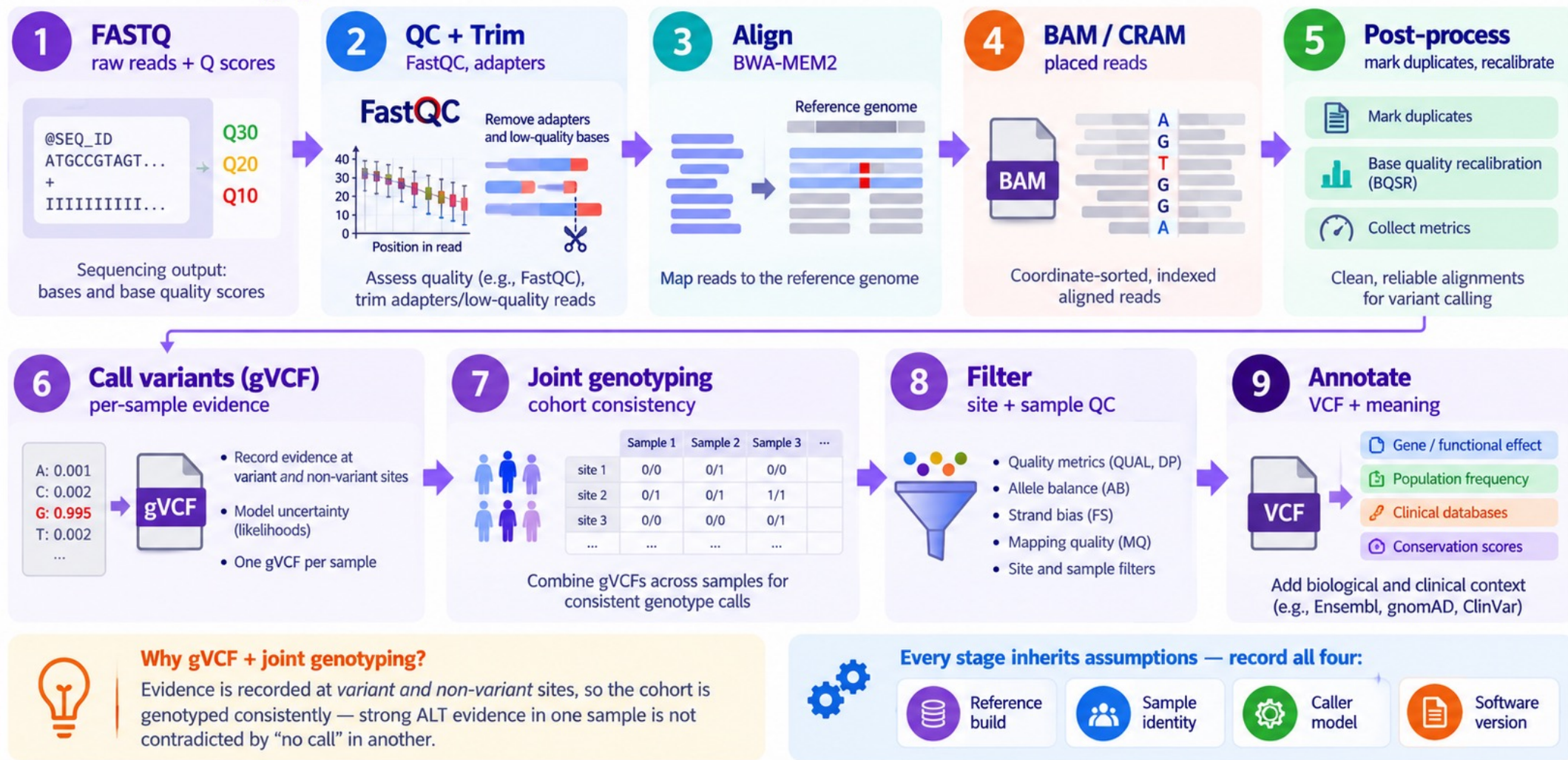
Widely used
Standard QC step in NGS analysis workflows



Tip

Run FastQC on raw reads before trimming and downstream analysis to identify potential issues.

The pipeline progressively transforms reads into testable variant hypotheses



Genomic file formats encode different objects — they are not interchangeable

File format	What it stores	Produced by	Key gotcha
 FASTA	Reference or consensus sequence	 Assembly / download	 No quality, no coordinates
 FASTQ	Raw reads + per-base qualities	 Sequencer base calling	 Unplaced — no position
 BAM / CRAM	Reads aligned to a reference	 Aligner (BWA-MEM2)	 Tied to one build + header
 VCF / gVCF	Variant sites + genotype evidence	 Variant caller	 Fields defined in its header
 BED	Intervals (regions)	 Analysis / annotation	 0-based, half-open
 GTF / GFF	Gene and transcript annotation	 Annotation databases	 1-based; version-specific

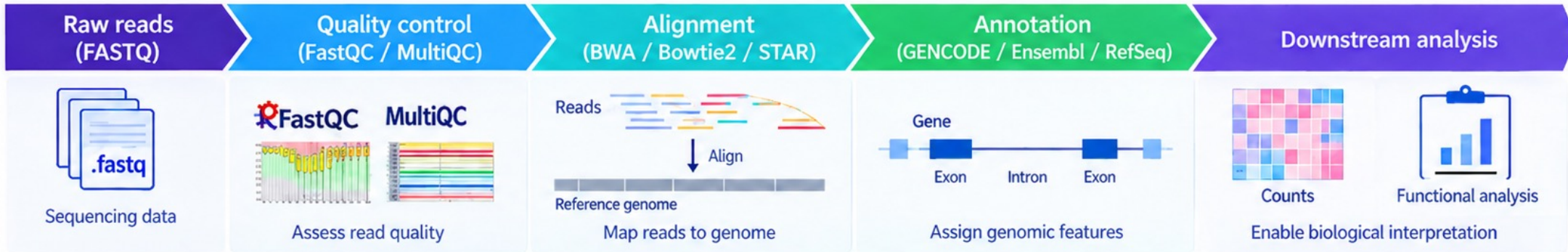


Provenance travels with every file: source, checksum, build, tool version, coordinate convention, sample mapping.

Alignment and Annotation: Key References and Resources

From raw reads to genomic features using trusted tools and databases

REFERENCE GENOME
GENE ANNOTATION
FUNCTIONAL INTERPRETATION
REPRODUCIBLE ANALYSIS



Reference Genomes



ENCODE-quality
gene annotation



Comprehensive
genome annotation



Curated reference
sequences

Common assemblies

Human: GRCh38 (hg38)

Mouse: GRCm39 (mm39)

Other: species-specific assemblies



UCSC Genome Browser
<https://genome.ucsc.edu>
Visualization and download

Alignment Tools

BWA

Burrows-Wheeler Aligner

Short reads (DNA-seq, ChIP-seq)
Li & Durbin, 2009
Bioinformatics



Fast and sensitive alignment
Langmead & Salzberg, 2012
Nat Methods

STAR

Spliced alignment (RNA-seq)

Dobin et al., 2013
Bioinformatics



Long reads (PacBio/ONT)

Li, 2018
Bioinformatics

Annotation Resources



- High-quality gene models
- Multiple transcript isoforms
- Regular updates
Harrow et al., 2012
Nucleic Acids Research



- Comprehensive annotation
- Comparative genomics
- Variation information
Yates et al., 2020
Nucleic Acids Research



- Curated, non-redundant
- Stable identifiers
- Widely used in pipelines
O'Leary et al., 2016
Nucleic Acids Research

Functional Annotation



- Protein function
- Domains and families
- Cross-database links
The UniProt Consortium, 2023
Nucleic Acids Research



- Pathway annotation
- Metabolic and signaling
- Disease relevance
Kanehisa et al., 2021
Nucleic Acids Research



- Gene Ontology terms
- Biological process
- Molecular function
- Cellular component
Ashburner et al., 2000
Nature Genetics

Best Practices



Use up-to-date
reference genomes
and annotations



Report tool versions
and parameters



Ensure reproducibility
with workflow management
(Snakemake / Nextflow)



Verify annotation sources
for your organism and question

Key Takeaway

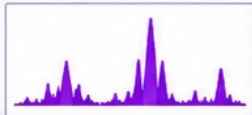


Accurate alignment and high-quality annotation are essential for reliable downstream analysis and biological discovery.

Regulatory genomics reads multiple layers of information from genomic DNA

Which proteins bind DNA?

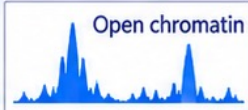
TF ChIP-seq / CUT&Tag / CUT&RUN



TF binding

Which DNA is accessible?

ATAC-seq / DNase-seq



Open chromatin

Which histone marks define chromatin state?

H3K27ac, H3K4me3, H3K27me3
ChIP-seq / CUT&Tag



H3K27ac (active)

H3K4me3 (promoter)

H3K27me3 (repressed)

Where does transcription start?

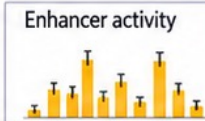
CAGE / RAMPAGE



TSS signal

What can the sequence do?

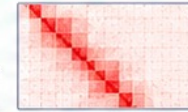
MPRA / STARR-seq



Enhancer activity

Which loci contact in 3D?

Hi-C / Micro-C / HiChIP / PLAC-seq



Which bases are chemically modified?

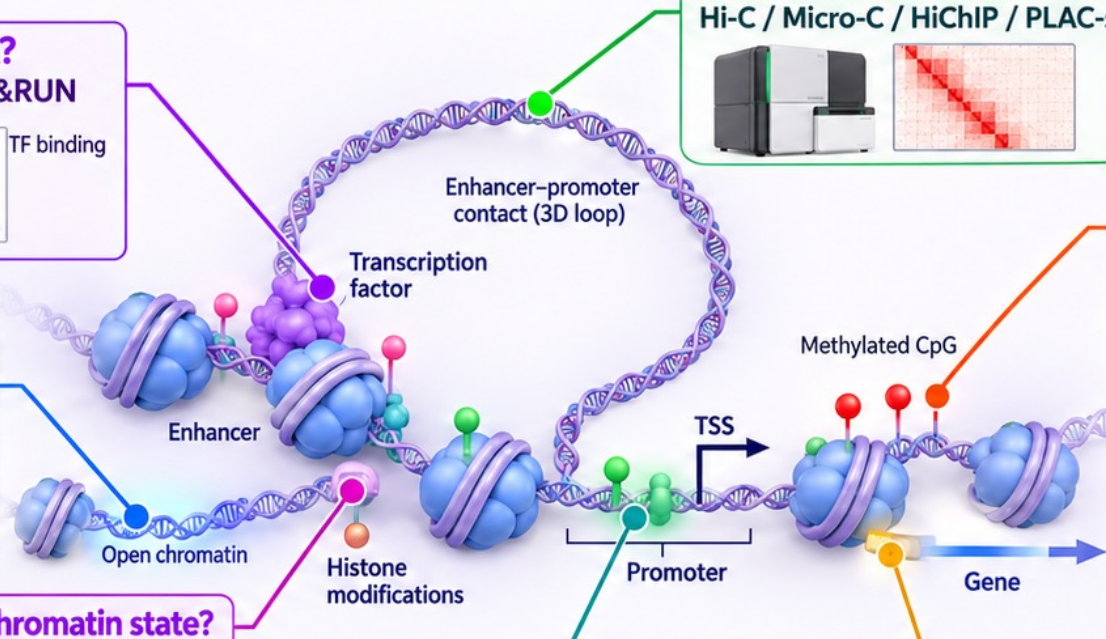
DNA methylation: WGBS / EM-seq



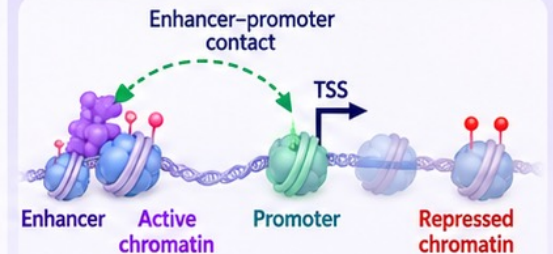
Methylated CpG



● Methylated CpG
○ Unmethylated CpG



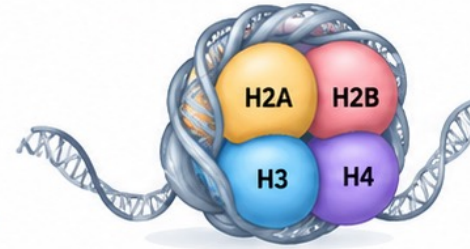
Integrative view: functional interpretation



No single assay defines function — integration across accessibility, occupancy, chromatin marks, methylation, and 3D contacts explains regulatory DNA.

Histone modifications and their major functions

A practical reference for chromatin-state interpretation in regulatory genomics



Nucleosome

- = activation / open chromatin
- = repression / silencing
- = DNA damage / cell cycle
- = structural or emerging regulatory roles

Acetylation – usually linked to open chromatin

H3K27ac	active enhancers and active promoters
H3K9ac	active promoters, open chromatin
H3K14ac	transcription activation
H3K18ac / H3K23ac	coactivator recruitment, active transcription
H3K56ac	nucleosome assembly, DNA replication, DNA repair
H4K16ac	chromatin decompaction, active chromatin

Methylation – context dependent

H3K4me3	active promoters / transcription start sites
H3K4me2	active or poised promoters
H3K4me1	enhancers (poised or active)
H3K36me3	gene bodies, elongation, splicing
H3K79me2/3	transcriptional activity
H3K27me3	Polycomb repression, developmental silencing
H3K9me3	constitutive heterochromatin, repeat silencing
H3K9me2	heterochromatin / gene repression
H4K20me1	chromatin compaction, cell-cycle control
H4K20me2/3	DNA damage response, heterochromatin

Phosphorylation – cell cycle and damage signaling

H3S10ph	mitosis, chromosome condensation, rapid gene activation
H3S28ph	mitosis and immediate-early responses
H3T3ph	centromere function and mitosis
γH2AX (H2AXS139ph)	DNA double-strand break response

Ubiquitination

H2AK119ub	Polycomb-mediated repression
H2BK120ub	transcription elongation; supports H3K4 and H3K79 methylation
H2AK13/15ub	DNA damage signaling and repair choice

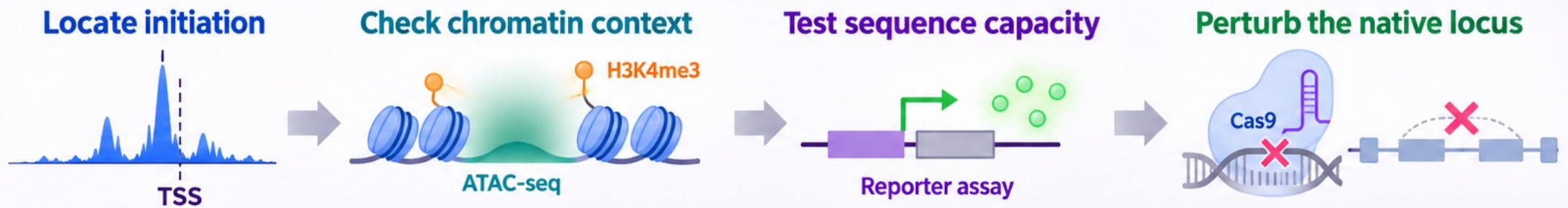
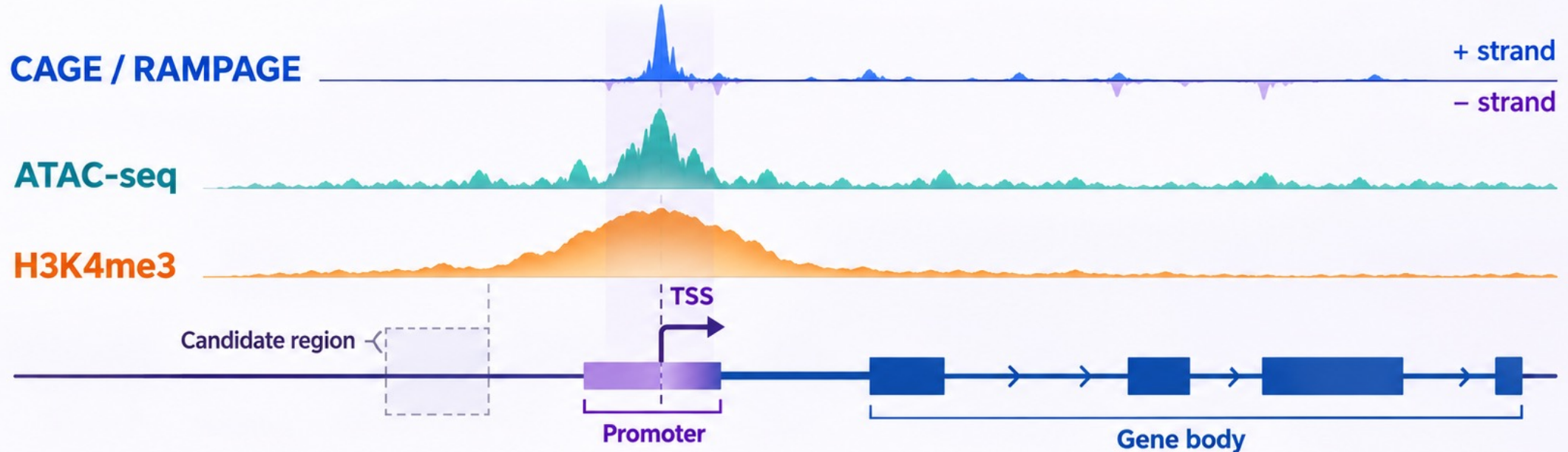
Other important PTMs

SUMOylation	transcription repression, genome stability, DNA repair
Citrullination	chromatin decondensation, inflammatory programs, NETosis
ADP-ribosylation	chromatin relaxation during DNA repair
Crotonylation	active transcription, enhancer and germ-cell programs
Lactylation	metabolism-linked gene activation
Propionylation / Butyrylation	active chromatin states
O-GlcNAcylation	transcriptional regulation and mitotic control

No single mark is sufficient: chromatin state is inferred by combining histone marks, DNA accessibility, TF binding, methylation, and 3D genome contacts.

Common assays: ChIP-seq, CUT&Tag, CUT&RUN, mass spectrometry

Active promoters require convergent evidence



Reporter activity shows capacity. Endogenous perturbation tests native function.

Core assays for promoter and regulatory DNA analysis

CAGE/RAMPAGE, ATAC-seq, and H3K4me3 provide complementary evidence in genomic omics

CAGE / RAMPAGE

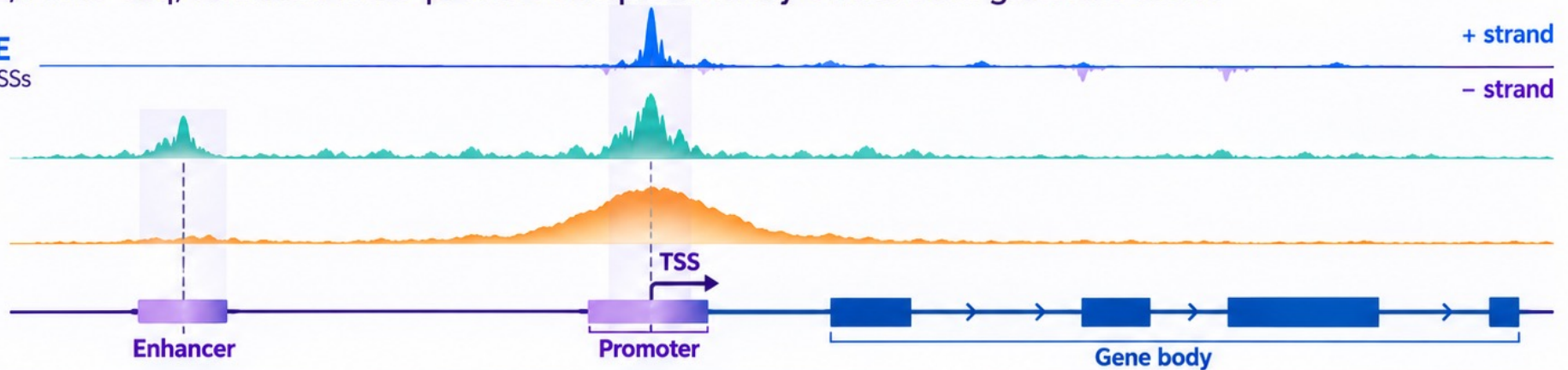
maps capped 5' ends / TSSs

ATAC-seq

open chromatin

H3K4me3

active promoter mark



CAGE / RAMPAGE



- **What it measures:** transcription initiation
- **Typical experiment:** capture 5' capped RNA ends
- **Best uses:** TSS mapping, promoter discovery, strand-specific initiation

ATAC-seq



- **What it measures:** chromatin accessibility
- **Typical experiment:** transposase inserts into open DNA
- **Best uses:** open chromatin, candidate enhancers, TF footprinting

H3K4me3 (ChIP-seq / CUT&Tag)



- **What it measures:** promoter-associated histone modification
- **Typical experiment:** enrich DNA bound to H3K4me3-marked nucleosomes
- **Best uses:** active promoters, chromatin state, regulatory annotation

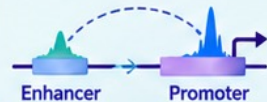
How they work together in genomic omics

Promoter annotation



Identify true TSSs and active promoters.

Enhancer-promoter logic



Accessible distal elements linked to gene control.

Cell-state comparison



Compare initiation and chromatin state across tissues or conditions.

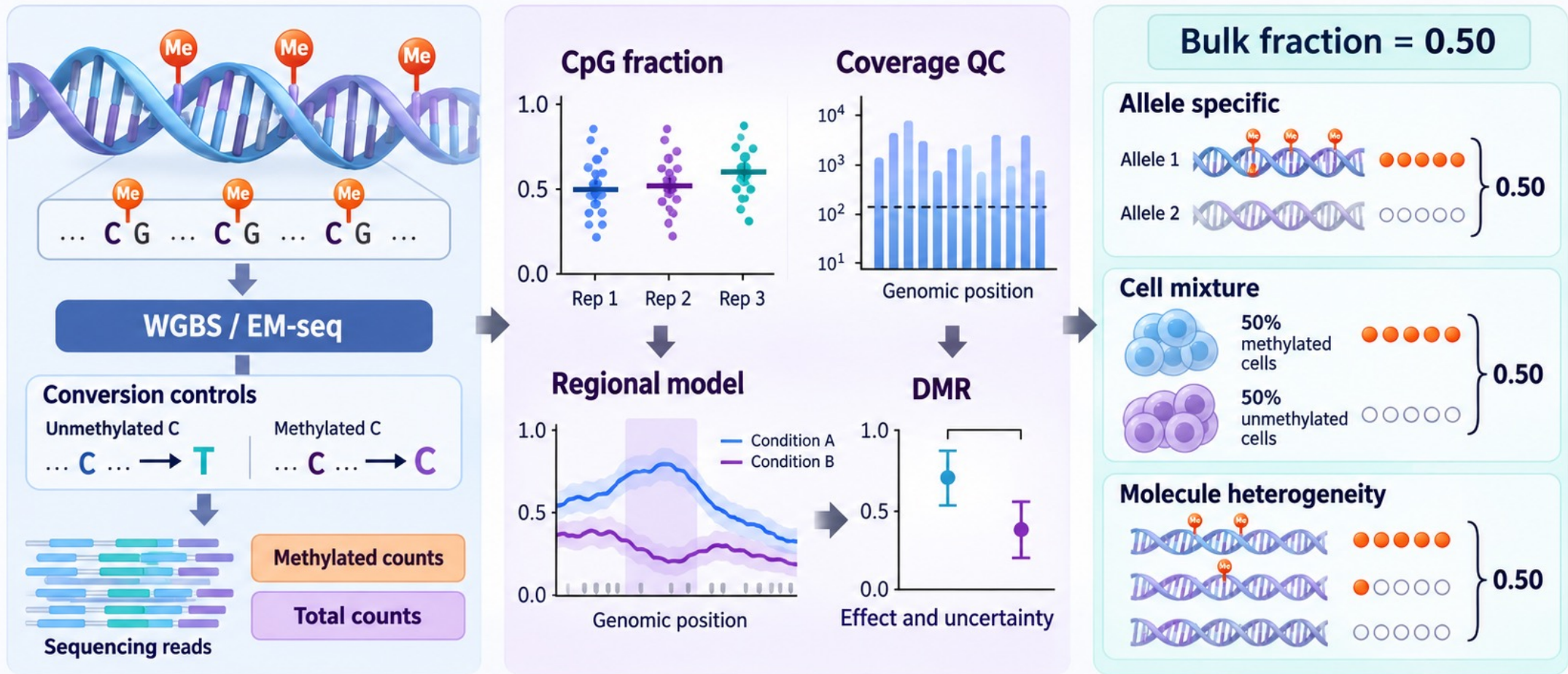
Disease variant interpretation



Prioritize noncoding variants in regulatory regions.

CAGE/RAMPAGE shows where transcription starts; **ATAC-seq** shows whether DNA is accessible; **H3K4me3** shows promoter chromatin state. Together they strengthen regulatory interpretation.

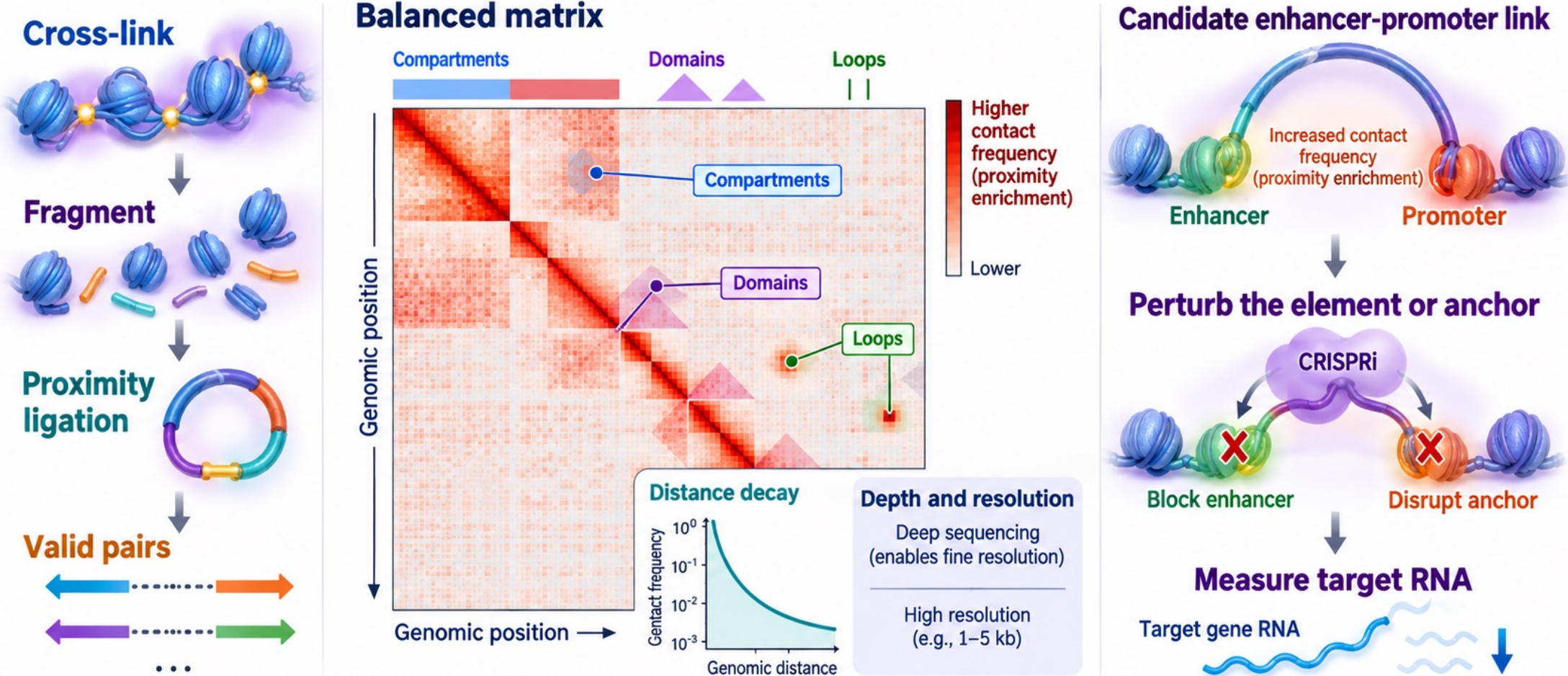
DNA methylation requires molecular resolution



A methylation fraction summarizes molecules, alleles, and cells.

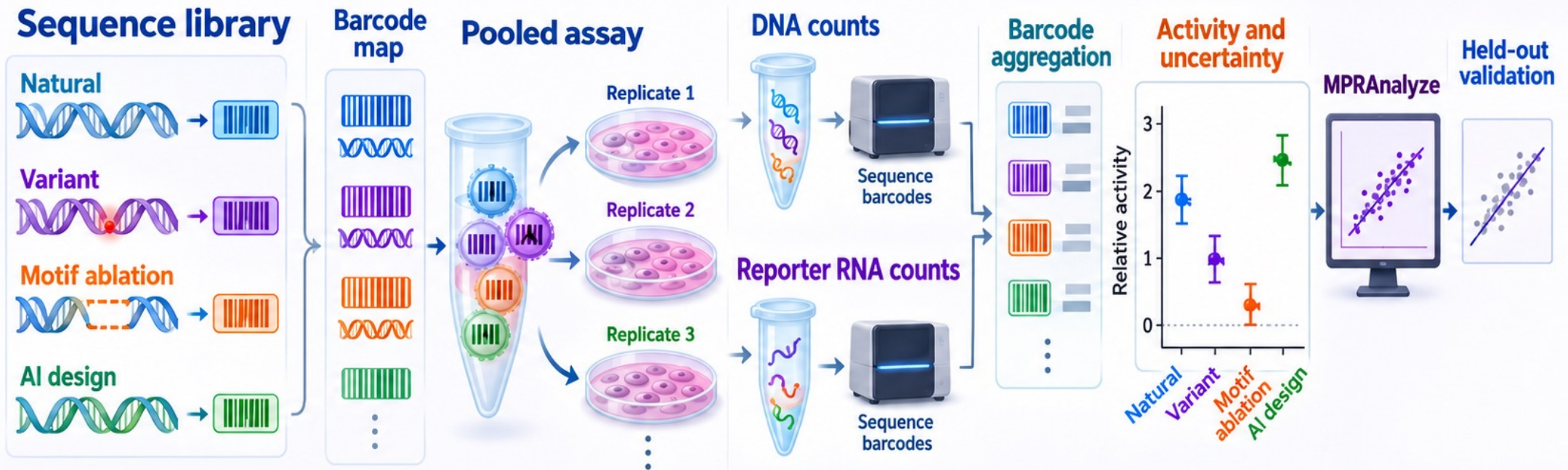
Standard WGBS / EM-seq: 5mC + 5hmC

Hi-C and Micro-C map genomic proximity



A loop prioritizes a regulatory link. Perturbation tests whether it controls expression.

MPRA measures sequence activity at scale



Normalized RNA relative to library DNA estimates sequence activity.



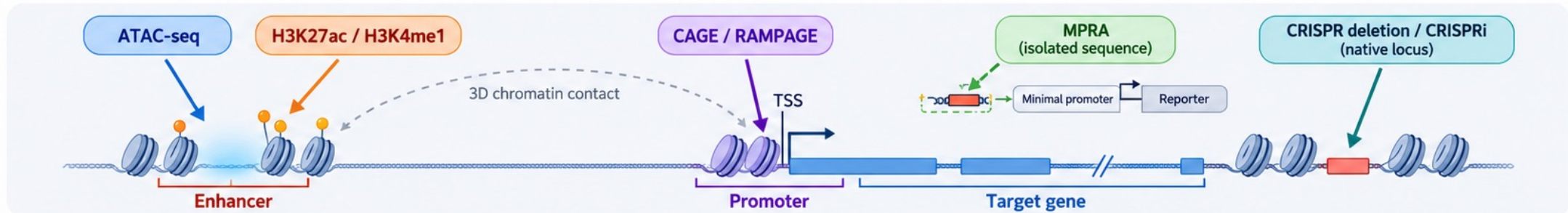
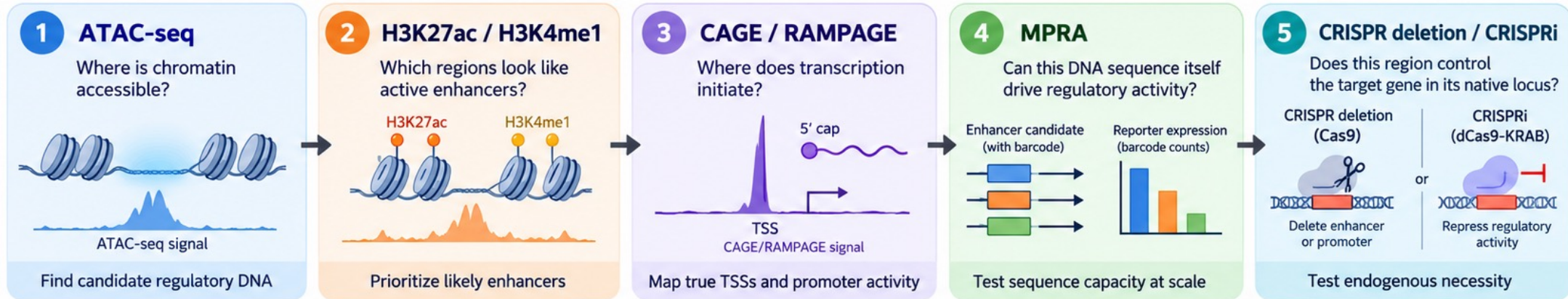
Barcodes are technical observations.



Biological samples define replication.

From regulatory clues to functional proof

How ATAC-seq, histone marks, CAGE/RAMPAGE, MPRA, and CRISPR answer different questions



Example workflow

- 1 ATAC-seq + H3K27ac → candidate enhancer
- 2 MPRA → enhancer capacity
- 3 CRISPR deletion / CRISPRi → endogenous target-gene control



Key interpretation

Observation ≠ function

Omics data nominate regions;
perturbation tests causality.

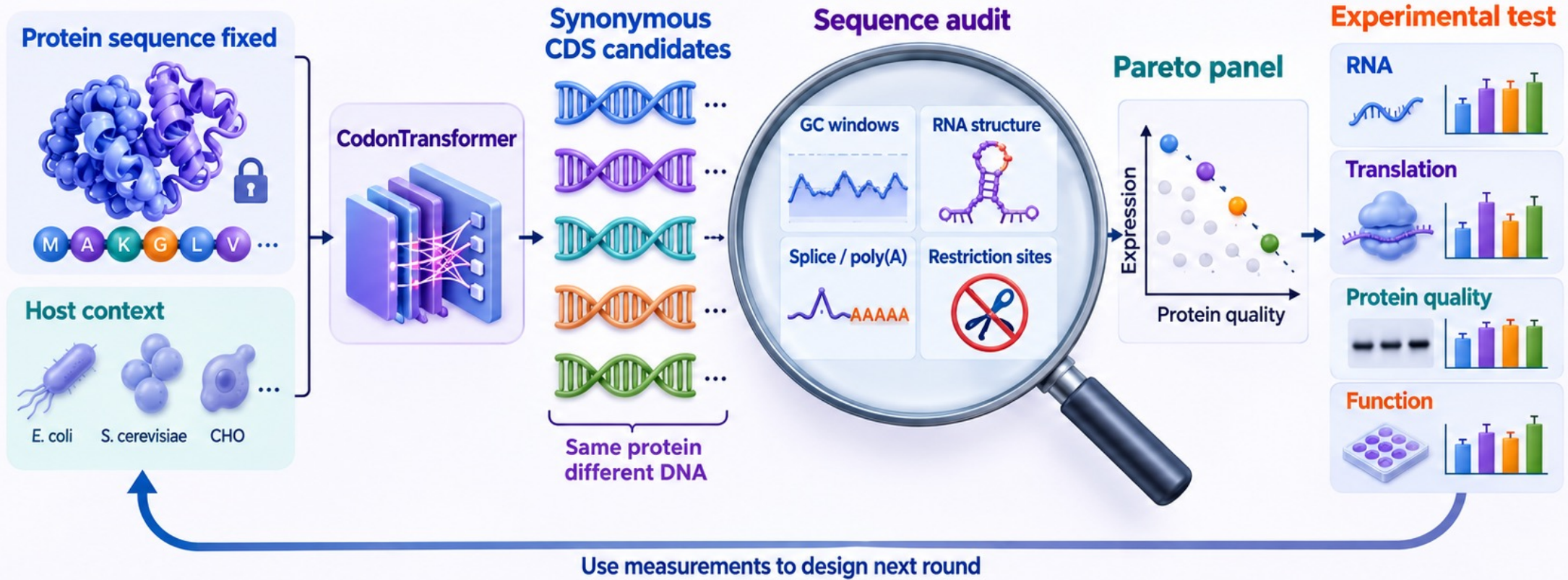


Important caution

**MPRA measures regulatory capacity,
not necessarily endogenous function.**

Chromatin context, 3D contacts, and native
genomic position still matter.

Codon optimization is a multi-objective design problem



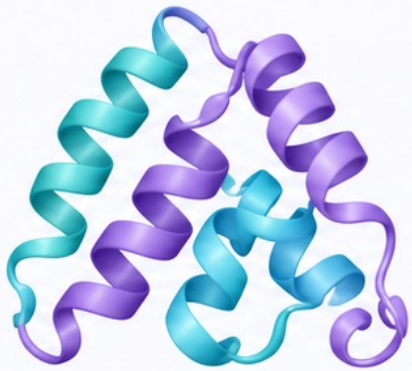
No single codon score predicts expression, folding, stability, and activity.

The protein sequence is fixed — we optimize the DNA

The same amino acid sequence can be encoded by many synonymous DNA sequences with different biological properties.



Protein (amino acid sequence)



M A K G L V ...

Amino acid sequence (fixed)

Protein structure & function
must be maintained



Genetic code is degenerate

Amino acid	3-letter (1-letter)	Synonymous codons (DNA, 5'→3')
Leucine	Leu (L)	TTA TTG CTT CTC CTA CTG
Serine	Ser (S)	TCT TCC TCA TCG AGT AGC
Arginine	Arg (R)	CGT CGC CGA CGG AGA AGG
Glycine	Gly (G)	GGT GGC GGA GGG
Alanine	Ala (A)	GCT GCC GCA GCG

Multiple codons encode
the same amino acid.



Example: Leucine codons

Leucine (Leu, L)

TTA

TTG

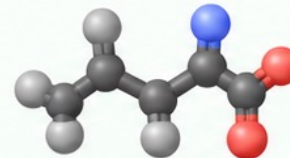
CTT

CTC

CTA

CTG

6 different
codons,
same amino acid.



Leucine
(C₆H₁₃NO₂)



A massive synonymous sequence space

$$\text{Number of possible sequences} = \prod_{i=1}^n k_i$$

where k_i is the number of synonymous
codons for amino acid i .

A protein of n amino acids can be
encoded by a huge number of
DNA sequences.

Example:

a 300 amino acid protein

$\sim 10^{180}$

possible DNA sequences

(assuming an average of 3 codons per amino acid).



Traditional codon optimization often chooses the most frequent codon.

Modern approaches ask: Which synonymous CDS gives the best biological performance?

AI explores the synonymous CDS space

AI models generate many coding-sequence candidates from the same protein and a chosen host context.

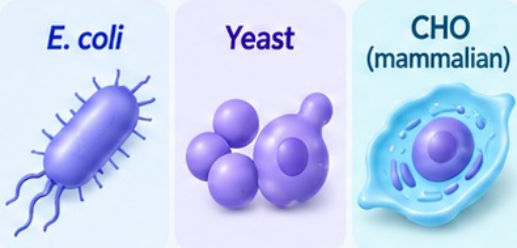
Inputs

Protein sequence



Keep the same amino-acid sequence

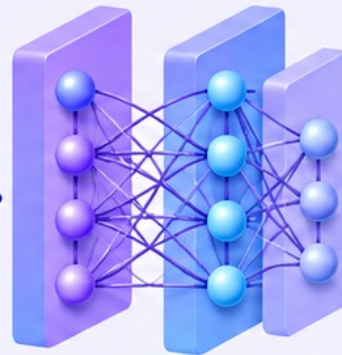
Host context



Choose the target host for expression

AI sequence generator

CodonTransformer /
generative model



Learn host-aware
coding patterns

Generated synonymous CDS candidates

Candidate 1

ATG GCT AAG GGC CTG GTG ...

Candidate 2

ATG GCC AAA GGT CTC GTG ...

Candidate 3

ATG GCA AAG GGC CTA GTT ...

Candidate 4

ATG GCG AAA GGT TTG GTA ...

Candidate 5

ATG GCT AAG GGT CTG GTT ...

...

Same protein, different DNA

First-pass scoring

Prioritize candidates before
deeper auditing



Codon usage

Match host preferences



GC%

Within optimal range



Motifs

Avoid unwanted motifs
(e.g., restriction sites)



Predicted expression

ML model score



Rank and select top candidates
for detailed evaluation



AI does not change the protein. It searches the vast synonymous DNA space more efficiently than manual design.

Candidate sequences must pass a sequence audit

AI-generated CDSs are filtered by biological and manufacturability constraints.

Candidate CDS sequences

Candidate 1

ATGCCACCGTAAACGCTG...

Candidate 2

ATGAGTCCGTTGACCTGAA...

Candidate 3

ATGCCCAAGGTTTACCGGAT...

Candidate 4

ATGGGCTCCTATGACCGTGA...

Candidate 5

ATGTTTCCCGGAATTCGAAA...

Candidate 6

ATGACCGTACCGTACCGTAC...

Sequence audit

GC windows



RNA structure



Avoid stable mRNA hairpins.

Splice / poly(A) sites

AG|GURAGU
AATAAA

Avoid cryptic processing signals.

Restriction sites

GAATTC



Avoid common restriction sites.

Repeats / homopolymers

AAAAAA TTTTT

Avoid long repeats or homopolymers.

Synthesis complexity



Reduce sequence liabilities.



Pass

Meets all biological and manufacturability criteria.

Candidate 1

ATGCCACCGTAAACGCTG...

Candidate 2

ATGAGTCCGTTGACCTGAA...

Candidate 6

ATGACCGTACCGTACCGTAC...



Flag / remove

Fails one or more criteria.

Candidate 3

ATGCCCAAGGTTTACCGGAT...

Issue: long homopolymer (TTTTT)

Candidate 4

ATGGGCTCCTATGACCGTGA...

Issue: predicted stable hairpin

Candidate 5

ATGTTTCCCGGAATTCGAAA...

Issue: restriction site (GAATTC)



Useful CDS design balances biological performance with practical synthesis and cloning constraints.

Codon optimization is a multi-objective problem

No single score captures expression, folding, stability, and activity.

Competing objectives



Expression

High protein yield



mRNA stability

Longer half-life, less degradation



Translation efficiency

Faster, more accurate translation



Protein quality

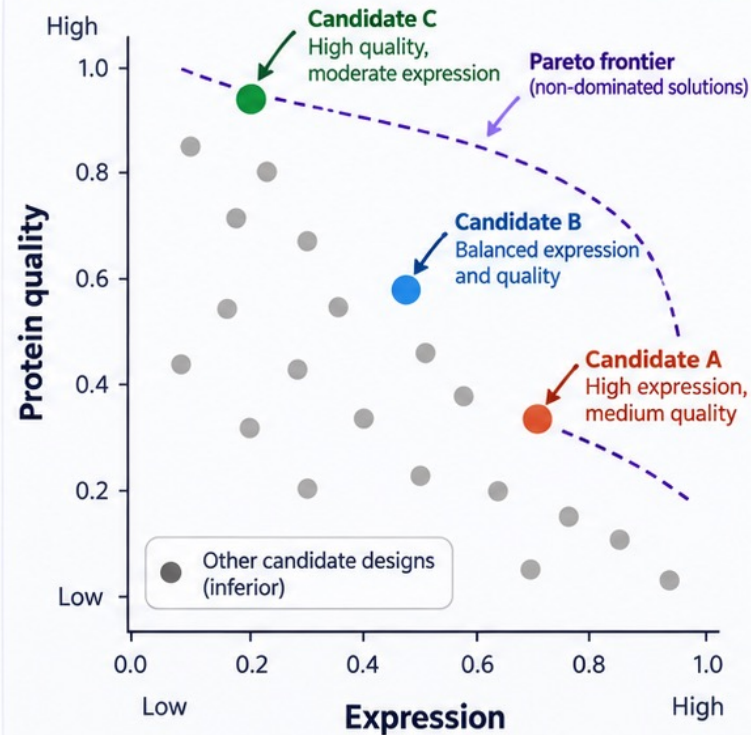
Correct folding, activity, low aggregation



Manufacturability

Robust production, scalable process.

Pareto panel



Choose according to project goals



Maximum yield

Top choice
Candidate A

Prioritize highest expression when yield is critical.



Best quality

Top choice
Candidate C

Prioritize protein quality, activity, and stability.



Balanced lead

Top choice
Candidate B

Choose a well-rounded candidate for overall performance.



Easy manufacturing

Top choice
Candidate D
(example)

Prioritize sequences with robust, scalable production.



Modern codon optimization selects trade-offs, not a single universally best sequence.

Experiments determine what really works

Wet-lab testing reveals which CDS designs improve real biological performance.

Candidate CDS designs from computational step

Candidate 1

ATG GCT AAG TTT GGA ...

Candidate 2

ATG GCC AAA TTC GGT ...

Candidate 3

ATG GCA AAG TTC GGC ...

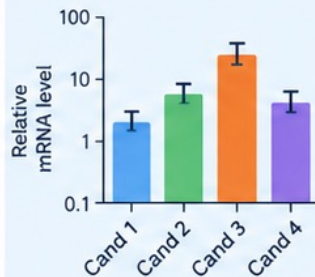
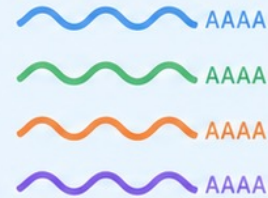
Candidate 4

ATG GCG AAA TTT GGA ...

Same protein,
different DNA sequences.

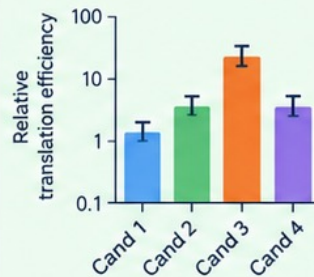
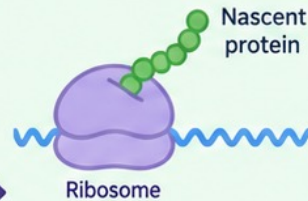
RNA abundance

Measure mRNA levels
(e.g., by RT-qPCR).



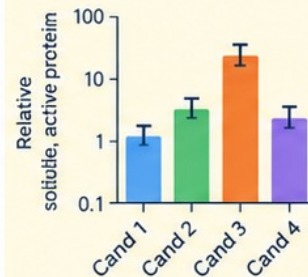
Translation efficiency

Measure protein synthesis
(e.g., ribosome profiling
or reporter assay).



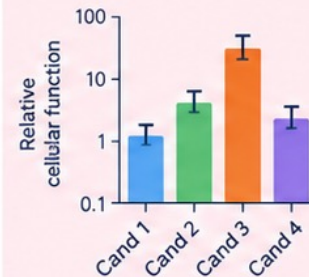
Protein quality

Assess correct folding
and stability
(e.g., solubility, activity).



Biological function

Measure the intended
cellular outcome
(e.g., enzyme activity,
cell growth, reporter signal).



Important replication rule

Biological samples define
replication.
Technical barcodes or
repeated reads are **not**
biological replicates.



Select the best performer

Candidate 3

ATG GCA AAG TTC GGC ...

Chosen based on real
biological performance.



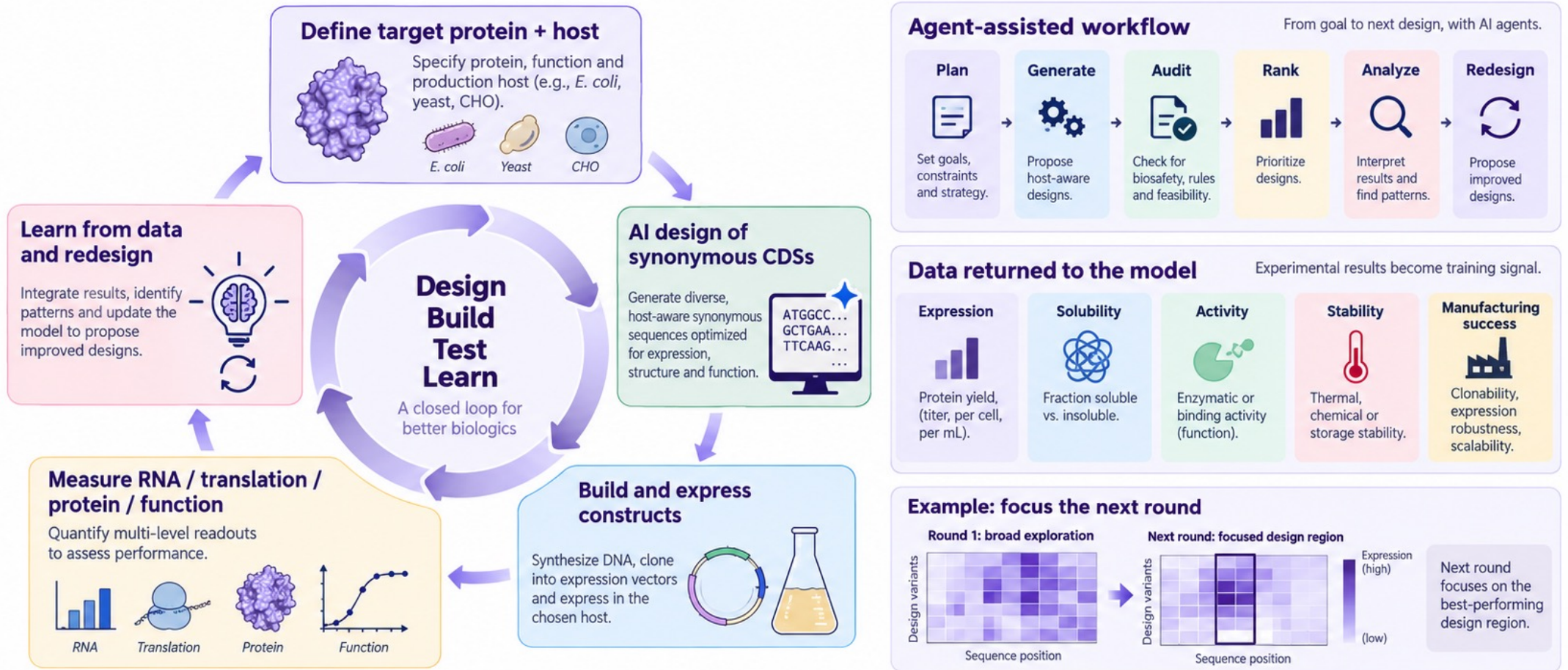
High RNA does not
always mean best
protein function.



AI proposes candidates, but experiments decide which codon design is best.

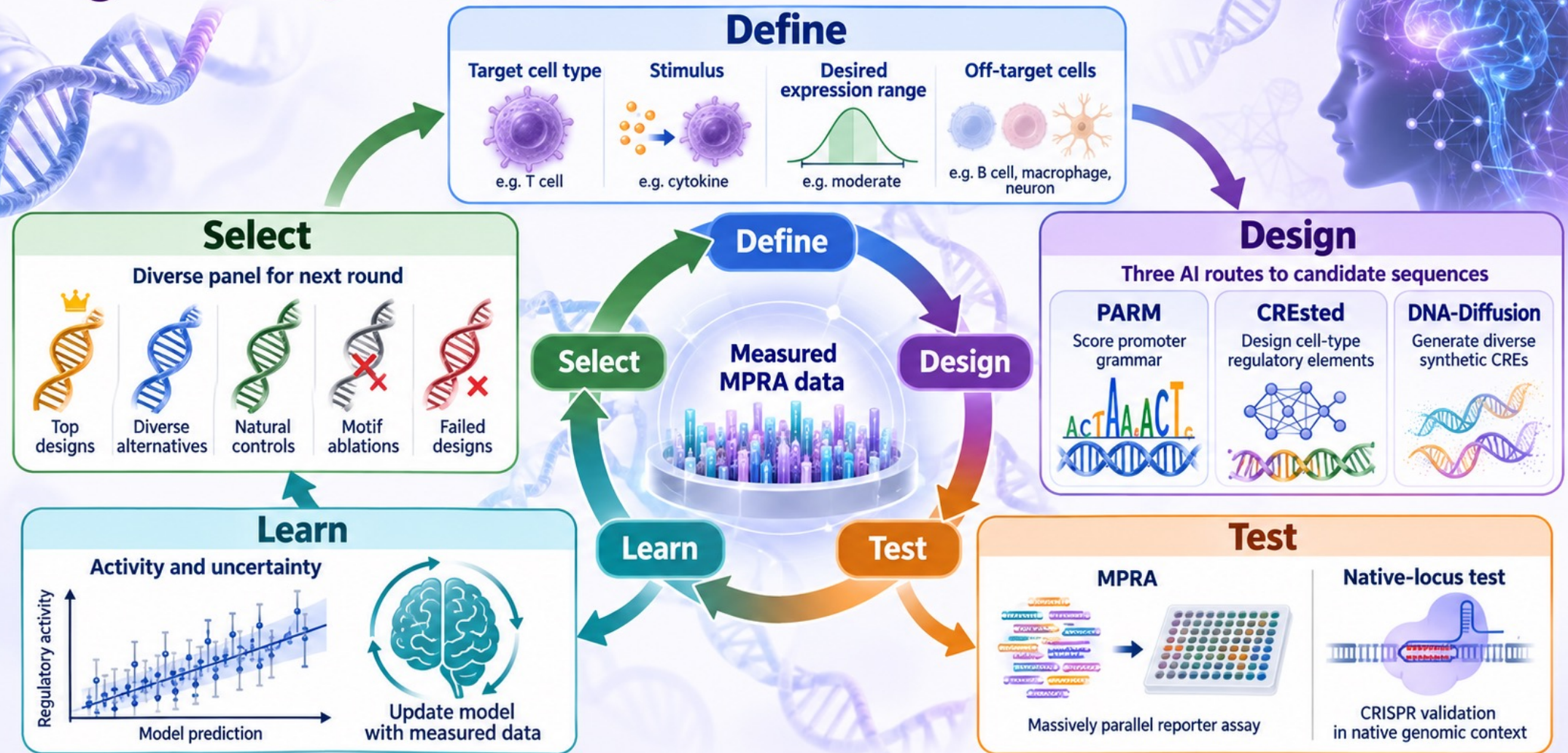
AI closes the design-build-test-learn loop

Experimental data feed back into the next round of host-aware sequence design.



The goal is not one round of optimization; it is a **closed-loop system** that improves with data.

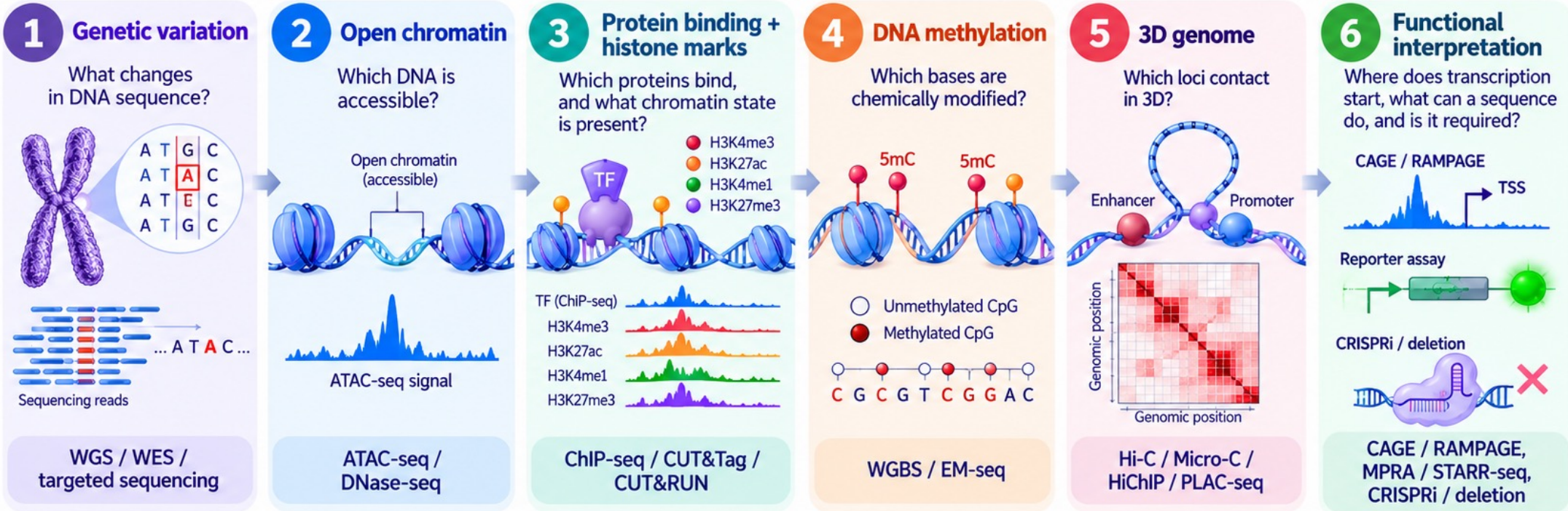
AI-guided regulatory design requires measured data



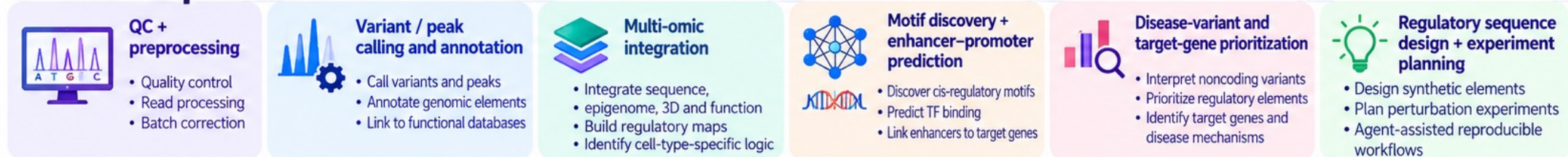
Model predictions propose sequences. Experiments determine regulatory function.

DNA-layer omics and AI: from sequence to regulatory function

Genomic, epigenomic, chromatin, and functional assays provide complementary evidence; AI helps analyze, integrate, predict, and design.



How AI helps



Omics nominate regions, AI integrates evidence, and perturbation tests function.

Overview: From FASTQ to Biological Insight

Common analysis workflow across genome-wide assays



Different assays, shared principles



ChIP-seq / CUT&Tag / CUT&Run



DNA methylation (WGBS/RRBS)



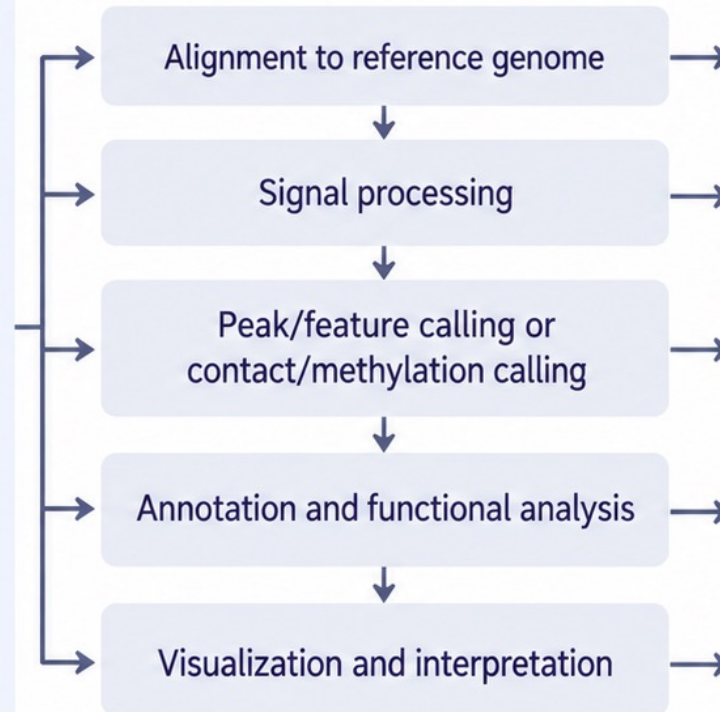
Hi-C / 3D genomics



ATAC-seq (for comparison)



RNA-seq (for comparison)



Biological insights



Regulatory elements



Epigenetic modifications



3D chromatin organization



Cell type / condition differences

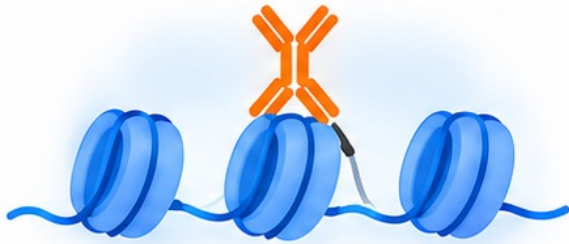


Mechanistic hypotheses

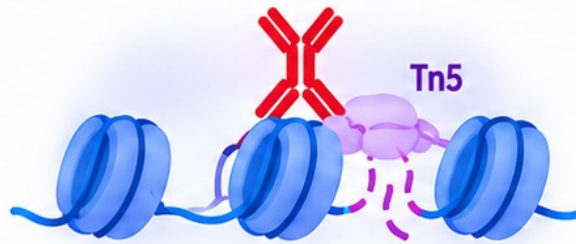
ChIP-seq, CUT&Tag and CUT&Run

Mapping protein–DNA interactions and regulatory elements

ChIP-seq
(antibody + IP)



CUT&Tag
(antibody + Tn5)



CUT&Run
(antibody + MNase)



FASTQ

Raw reads



QC

Quality control
FastQC/MultiQC



Alignment
(Bowtie2/BWA)

Map to genome



Peak calling
(MACS2)

Identify
enriched regions



Annotation
(HOMER/ChIPseeker)

Link to genes
and functional elements



Visualization

Genome browser
and motif analysis

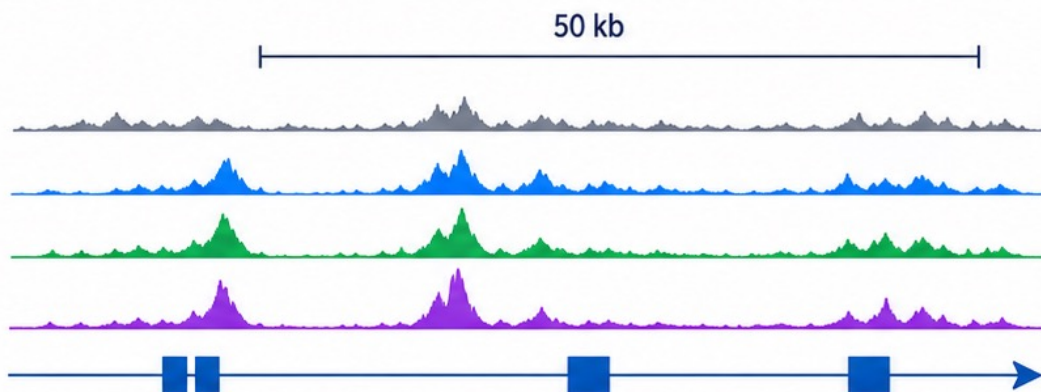
Input

ChIP-seq (H3K27ac)

CUT&Tag (TF)

CUT&Run (TF)

Gene



Typical downstream analyses

-  Peak annotation (promoter, enhancer)
-  Motif analysis
-  Integration with RNA-seq/ATAC-seq
-  Differential binding analysis
-  Pathway and gene regulation analysis

From Peaks to Function: MACS2, HOMER and ChIPseeker

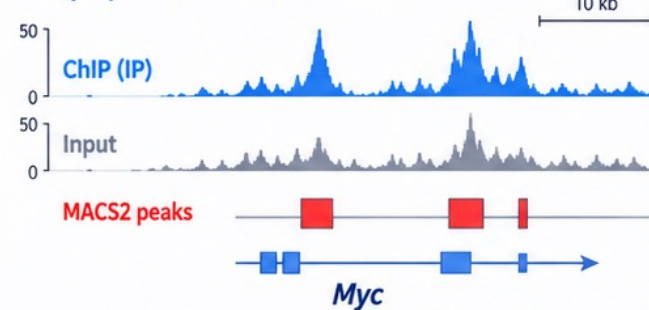
A comprehensive analysis pipeline for ChIP-seq data

PEAK CALLING
MOTIF DISCOVERY
GENOMIC ANNOTATION
BIOLOGICAL INTERPRETATION

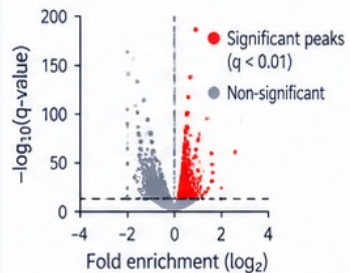
1. MACS2 – Peak calling

Identify enriched regions from ChIP-seq data

Example peaks (IGV view)



Peak significance



MACS2 output (narrowPeak)

chr	start	end	q-value	signal
chr1	345672	345921	1e-50	120.3
chr1	678912	679213	2e-25	86.7
chr2	123456	123789	4e-18	54.1
chr3	987654	987920	6e-12	32.5
...	...	...	...	...

Key points (MACS2)

- Identify high-confidence peaks ($q\text{-value} < 0.01$)
- Provide peak location, signal strength and enrichment
- Output:** narrowPeak / broadPeak files for downstream analysis

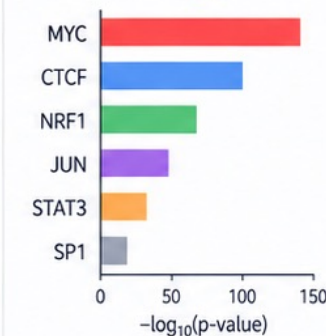
2. HOMER – Motif discovery and peak annotation

Find enriched motifs and assign genomic features

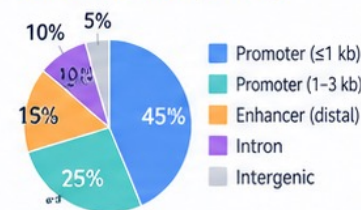
De novo motif discovery

Motif	Best match	p-value
CACGTG	E-box (MYC)	1e-120
GGAATG	ETS (ELF1)	1e-45
TTGACT	AP-1 (JUN)	1e-28
CCCTC	CTCF	1e-20
TGACA	NRF1	1e-15

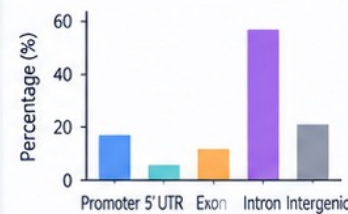
Known motif analysis



Peak annotation (HOMER)



Genomic distribution



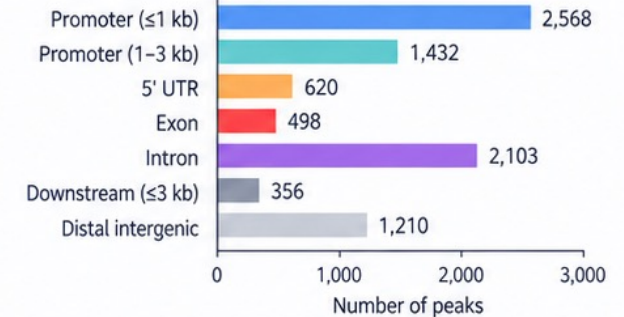
Key points (HOMER)

- Discover overrepresented DNA motifs (*de novo* and known)
- Annotate peaks to genomic features (promoter, enhancer, intron, etc.)
- Provide biological interpretation (transcription factors and regulatory elements)

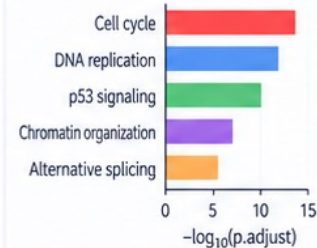
3. ChIPseeker – Functional annotation

Link peaks to genes and pathways

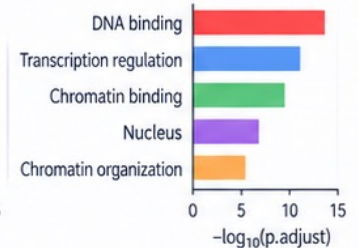
Annotation to nearest genes



Top enriched pathways (KEGG)



Gene ontology (GO)



Key points (ChIPseeker)

- Annotate peaks to nearest genes and genomic features
- Perform GO/KEGG pathway enrichment analysis
- Link ChIP-seq signals to biological functions and pathways

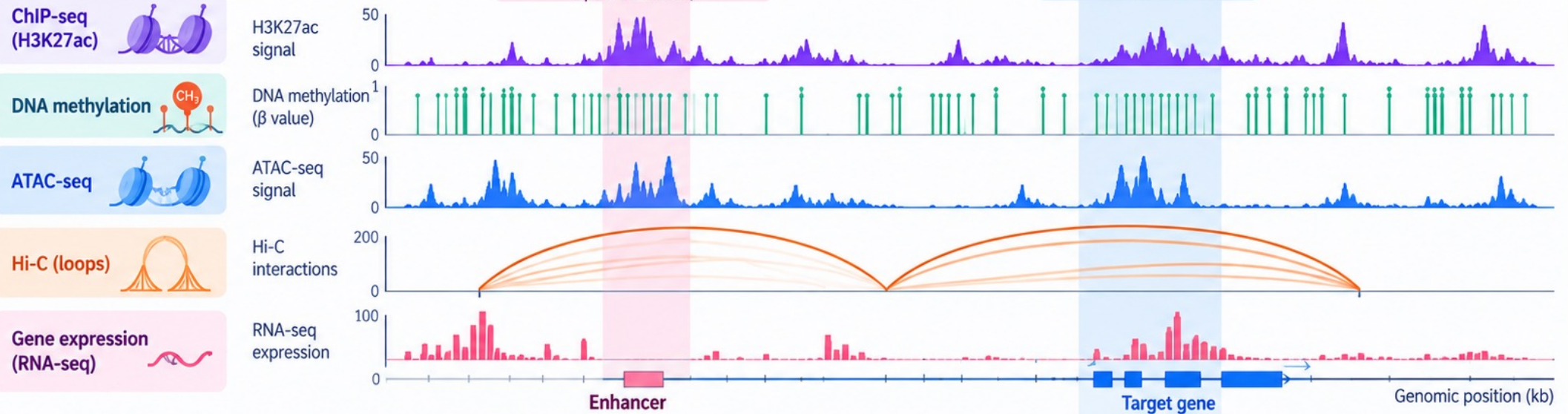


From peaks to pathways: MACS2 finds the signal, HOMER reveals the motifs, and ChIPseeker connects them to genes and biological functions.

Integrative Analysis Across Epigenomics

A multi-omics view of genome regulation

MULTI-OMICS
SYSTEMS BIOLOGY
GENE REGULATION
PRECISION MEDICINE



Integrative questions



Which regulatory elements control gene expression?



How do epigenetic modifications shape chromatin architecture?



How do these layers change across cell types or diseases?



Can multi-omics integration identify novel therapeutic targets?

Key insights from integration



Link enhancers to target genes



Reveal coordinated epigenetic changes



Identify cell type-specific regulatory programs



Discover disease-associated regulatory elements



Prioritize therapeutic targets

Common tools



IGV / Juicebox (visualization)



bedtools / deepTools (signal analysis)



HiC-Pro / Juicer (contact analysis)



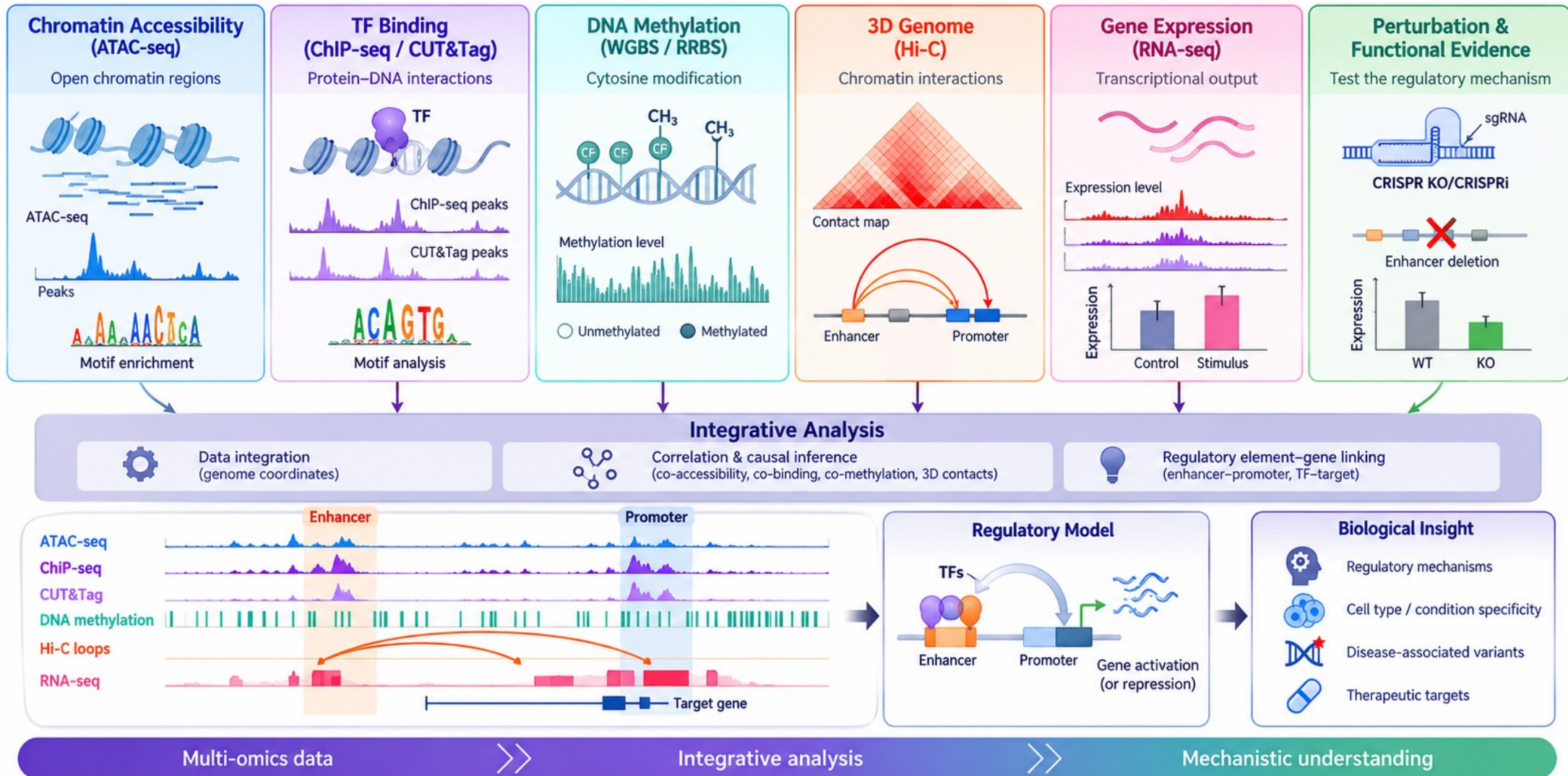
Enrichr / GREAT (functional annotation)

Multiple epigenomic layers work together to control gene regulation.

Integrative Epigenomic Analysis: From Multiple Layers to Regulatory Mechanisms

Combine chromatin accessibility, TF binding, DNA methylation, 3D genome and gene expression to decode gene regulation

MULTI-OMICS
MULTI-SCALE
BIOLOGICAL INSIGHT



A plan-first prompt produces safer R code for variant prioritization

Better prompts
Better science

VARIANTS
ANALYSIS
IMPACT



Weak prompt

Vague → Risky code

"Write R code to filter my variant table to pathogenic candidates."



No reference build specified
(e.g., GRCh37 or GRCh38?)



No schema or column names
(what fields are available?)



No quality checks or filtering logic
(risk of incorrect or unsafe code)



Overclaimed output
"pathogenic candidates" is too strong and may be misleading



Strong prompt — plan first

Specific → Safer, reliable code

1

Provide context and schema

Variant table				
CHROM	POS	REF	ALT	...
1	12345	A	G	...
2	67890	C	T	...

"Context: **GRCh38** diploid germline variant table.

Columns: **CHROM**, **POS**, **REF**, **ALT**, **FILTER**, **DP**, **GQ**, **gnomAD_AF**, **ClinVar_CLNSIG**.

2

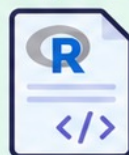
Ask for a plan before code



- Before coding, list assumptions.
- Include schema checks, filtering logic, outputs, and a verification checklist.

3

Request safe, well-documented R code



- Write commented R code that **preserves the raw file**.
- Treat missing frequency as **unknown**.
- **Report row counts at every filter**.
- Save **sessionInfo()**.
- State clearly that the result is **not a diagnosis**.



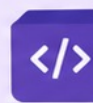
Plan

Define context, schema, and analysis strategy



Approve strategy

You review and confirm before any code exists



Generate R code

Model writes commented, reproducible code



Check outputs

Verify results, logs, and sessionInfo()



You approve the strategy before any code exists — the model accelerates execution, not decisions.



The plan also becomes documentation — paste it into your verification log next to sessionInfo().

Transparent
Reproducible
Safer Science

Local analysis combines Linux, R, and AI

Raw files
(from sequencer)

sample1.fastq.gz

sample2.fastq.gz

sample3.fastq.gz

>_ Linux tools

FastQC / MultiQC

```
fastqc sample1.fastq.gz
fastqc sample2.fastq.gz
multiqc .
```

QC summary



samtools

```
samtools view -bs aln.sam > aln.bam
samtools flagstat aln.bam
```

BAM checks

Mapped: 98.7%
Proper pairs: 97.1%
No. reads: 48,231,902

bcftools

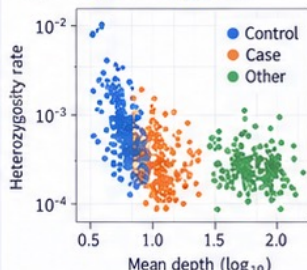
```
bcftools mpileup -f ref.fa aln.bam \
| bcftools call -m -Oz -o cohort.vcf.gz
bcftools norm -f ref.fa cohort.vcf.gz
-Oz -o cohort.norm.vcf.gz
```

Normalized VCF

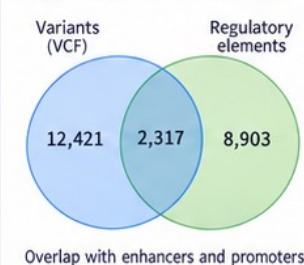
Variants: 4,982,311
SNPs: 4,312,660
Indels: 669,651

R / Bioconductor

Cohort QC with ggplot2



Regulatory interval overlaps



```
library(VariantAnnotation)
library(GenomicRanges)
library(ggplot2)
# read and annotate VCF
# overlap with regulatory intervals
# generate QC plots
```

AI assistant



Draft code

Suggest commands and R code



Explain errors

Diagnose and propose fixes



Summarize QC

Turn results into a clear report



Human verification

- ✓ Check commands
- ✓ Verify parameters
- ✓ Validate biology
- ✓ Confirm conclusions

Genomics

Bioinformatics

R for Genomics

Scientific Computing

My
Analysis
Workstation

Local
Data
Local
Control

Same Commands
Same Results
Trust the Science.



Guardrails protect raw data

- ✓ data_raw is read-only
- ✓ All steps are logged
- ✓ Versions recorded
- ✓ Checksums tracked
- ✓ Reproducible workflow

Reproduce
Validate
Discover



AI accelerates analysis. The analyst controls identity, reference, evidence, and interpretation.

Summary

From DNA to discovery — multi-omics, AI, and reproducible analysis

Data → Knowledge → Impact

1. DNA-based Omics

Different assays, complementary views of the genome



Whole Genome Sequencing (WGS)

Complete variation landscape



Whole Exome Sequencing (WES)

Coding variation



Targeted Sequencing

Focused regions/genes



Epigenomics (ChIP-seq, ATAC-seq)

Regulatory elements



3D Genomics (Hi-C)

Genome organization



Single-cell Genomics

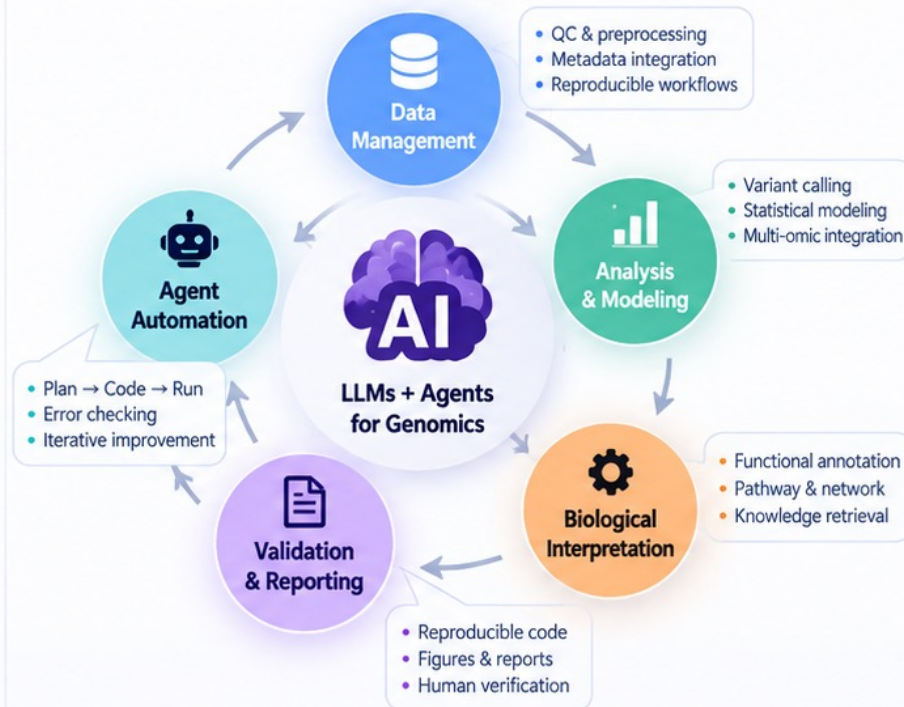
Cell-type resolution

...

More emerging technologies

2. AI-Enabled Analysis

From raw data to interpretable biology



3. Key Takeaways



Multiple DNA-based omics

provide complementary views from variation to regulation and 3D organization.



AI and AI agents

accelerate data analysis, integration, and hypothesis generation.



Plan first, execute reproducibly

Clear questions, rigorous design, and well-documented workflows are essential.



Interpretation turns data into discovery

Integrate evidence, validate findings, and consider biological context.



Human-in-the-loop

AI is a powerful assistant, but final interpretation and decisions require expertise.

From genomes to better lives

LEARN | ANALYZE | DISCOVER | MAKE AN IMPACT

